ProveIt research memorandum

Marked Parallelity Interfaces and Certified Handlebody Cores

Polynomial local topology and exact sector decomposition for the unknot-recognition programme

Prepared for Vladimir Reshetnikov and the ProveIt project
Research and implementation developed with ChatGPT

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Abstract

The present contribution addresses two explicit bottlenecks in compressed unknot recognition. First, a normal cut already has a source-certified parallelity bundle and a complementary core of at most $30T$ handles, but component profiles alone do not determine its attaching surfaces. We recover cyclic boundary words, old-boundary gaps, and fibre transitions using two additional compressed orbit queries. Maximal proper runs of core ports are disk interfaces; a complete marked boundary circle is an annular interface. We combine this information with geometric co-core slides and raw handle cancellation. Saturating occurrence counts at two preserves every admissible raw pivot. In the square, genus-zero case, a source leaf order is complete for the entire raw cancellation class, and an explicit induction proves that the exceptional core in every Fibonacci layered-torus fixture is a ball. Successful certificates give the exact handlebody census of the whole supplied cut, including binary multiplicities, without constructing its individual pieces.

Second, we split a supplied quadrilateral-sector matching space into its finest coordinate factors. Although canonical triangle minima can mix those factors, nonnegative vertex-link Euler characteristics force the global normalized Euler maximum to occur in a single factor. Complete standard-ray enumeration requires a coarser interaction decomposition. A genuine solid-torus counterexample separates these two statements. Another genuine family reduces the maintained global subset iterator from exponentially many candidate solves to linearly many local solves, with polynomial source preparation and verification.

The deliverable includes additive Python implementations, separate checkers, exact source fixtures, adversarial tests, independent Regina comparisons, and reproduction scripts. These are local algorithmic and geometric results. They do not establish a quasi-polynomial bound for the complete recognizer: useful-surface coverage, patterned-state transport, and global search bounds remain explicit research obligations.

Audited source. https://github.com/VladimirReshetnikov/ProveIt
`d2c5e1707fe783bbed06743ac732d70d83e6f27b`

Delivery status. Additive research contribution for review and integration; no repository publication is part of this delivery.

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1 Research position and scope

The appropriate next step is to make compressed geometric information usable by an exact topological consumer. The audited repository already counts and classifies normal-surface components, reconstructs actual cut cells and their boundaries, and extracts the full source parallelity bundle in handle indices zero, one, and two. It also retains the geometrically attached exceptional core. Replacing that construction with a prism-only approximation would lose information and undo an improvement already present in the source.

The missing information addressed here is smaller and more specific. A list of ports belonging to a base component gives membership, but it does not say which ports are consecutive, whether old boundary separates them, or whether they occupy a complete boundary circle. Those distinctions decide whether the interface with the core is a disk or an annulus. Once they are available, the existing geometric core can support a conservative, independently checked handlebody classifier. A separate source-coordinate decomposition reduces some of the search needed to find candidate surfaces in the first place.

1.1 What is new here, what is inherited, and what is open

Topic	Precise status in this contribution	Classification
Binary orbit computation	Polynomial weighted AHT and the maintained suffix-fold bound are used with attribution.	Inherited
Source parallelity	Complete bases, fibre charts, at most $30T$ core handles and $66T$ ports already exist.	Inherited
Marked cyclic interface	Two source-derived queries recover maximal attachment intervals and full marked circles, with independent replay.	New operator and proofs
Geometric core cancellation	Raw co-core counts admit exact saturation; source leaf elimination becomes a certified geometric ball test.	New geometric application
Layered cores	An explicit all-size source calculation gives a ball core for every positive multiple of the Fibonacci fixture.	New family theorem
Sector factors	Finest matching blocks, Euler localization, standard interaction clusters, and a geometric separation family.	New source-level development
Complete recognition	No bound on useful candidate coverage, all inherited patterns, or all hierarchy states is proved.	Open

Table 1: The novelty claim is relative to the pinned ProveIt source. Classical linear algebra, handle theory, and orbit algorithms are not claimed as new.

The permanent-one criterion for raw positive elimination already appears in report 56. Its new role here is geometric: raw occurrences are authenticated intersections with a co-core system, and the positive update is realized by actual disk slides. An abstract presentation simplifier is not substituted for a three-manifold proof. Likewise, the polynomial control of dense weight runs is already proved in `synthesis/suffix_folds.tex`; the sharper sparse-mark bound and explicit selector accounting below assemble the bounds needed by the new interface, rather than purporting to discover weighted AHT.

1.2 Relationship to the quasi-polynomial programme

We use *quasi-polynomial* in its usual broad sense $\exp((\log n)^{O(1)})$. The stronger target $n^{O(\log n)}$ is $\exp(O((\log n)^2))$; these formulations should not be conflated. Lackenby's March 2021 lecture handout announces $2^{O((\log n)^3)}$ for diagrams with n crossings[2]. The July 2026 manuscript

gives a new hierarchy-based recognition algorithm; its Section 9 explicitly separates iteration counting from an analysis of all step costs[3]. We do not infer a checked implementation or a complete complexity proof for ProveIt from an announced bound.

Our local bit-polynomial results are compatible with either target. They eliminate dependence on the *numerical multiplicity* of a supplied cut in the accepted class, and the sector result replaces total matching dimension by local block dimensions when splitting occurs. Neither statement bounds the number of candidates or the number and size of future patterned states. The difference is consequential: polynomial verification of exponentially many candidate certificates still permits exponential search.

2 Input model and the authenticated geometric interface

The runtime contract is a finite, connected, compact, orientable triangulated three-manifold M with one torus boundary, supplied by source face pairings, and an admissible integral standard normal vector $x \in \mathbb{N}^{7T}$. The surface $S(x)$ may be empty, disconnected, or one-sided. The source validator checks the relevant manifold links, face inverses, matching equations, and quadrilateral constraints. The exact cut is denoted $M \setminus\setminus S$: equivalently, remove the interior of a sufficiently small regular neighbourhood and round its corners. One-sided surfaces use their actual normal interval bundle, not two fictitious orientable copies.

Put

$$D = \sum_i x_i, \qquad B = 1 + \lceil \log_2(D+1) \rceil.$$

The coordinates have $O(TB)$ bits, and the face pairing has $O(T \log(T+1))$ bits. We charge arithmetic on binary endpoints and weights; an operation on a B-bit integer is not assigned constant cost. The article uses n only for original diagram size and T for the triangulation under discussion. No unproved bound relating a later hierarchy triangulation to n is hidden by switching those parameters.

2.1 The inherited source theorem

The following is the precise established input to the new geometric arguments. It is proved in the audited source's parallelity chapter and its raw cut-cell construction[6].

Proposition 2.1 (Source parallelity contract, inherited)**.** *The supplied cut decomposes into an embedded interval bundle $\mathcal{B}$ over a possibly disconnected compact surface B_0 and a complementary geometric handle core K. There is a marked subset $A \subseteq \partial B_0$ such that*

$$K \cap \mathcal{B} = p^{-1}(A), \qquad (h_0(K), h_1(K), h_2(K), h_3(K)) \le (6T, 8T, 12T, 4T)$$

coordinatewise. The packet has $O(T)$ source charts, including the complete core attaching cycles, fibre reversals, and at most $P \le 66T$ singleton core ports. Its base representation has at most $17T$ polygon families, $44T$ gluing families, and $68T$ side slots, with binary rank intervals.

The selected handles are the chambers between consecutive identical normal disks, the internal-face strips between consecutive parallel arcs, and the gaps between consecutive intersections on an interior edge. The complete base uses truncated triangles and quadrilaterals, rectangles, and edge-link polygons. Its gluing includes every selected zero–one, zero–two and one–two contact. Omitting the last two kinds would invalidate the embedding and intersection assertion. The exceptional lower-index attachments stay entirely in K; they are not group relators detached from source charts.

Four existing compressed queries determine the base-component profiles, actual boundary counts, orientability, and port membership. They also check

$$\chi(M \setminus\setminus S) = \chi(K) + \chi(B_0) - \chi(A). \tag{1}$$

A base profile containing any port has multiplicity one. Unported components may have large binary multiplicity. This distinction is necessary when a finite attachment graph is formed: it is incorrect to turn a profile of multiplicity m into a single ordinary vertex.

2.2 Trust boundary and certificate meaning

The source-level geometric result in Proposition 2.1 is a mathematical dependency. The existing checker reconstructs its charts from the raw cut-cell reference model using a different extraction route. The new checkers authenticate their inputs through that checker. Thus an accepted certificate depends on the source theorem and the correctness of the small verifier stack, as any certified numerical implementation depends on its trusted primitives. It is not a formal proof in Lean, Coq, or Isabelle.

The standalone core helper has a narrower contract: its caller must already have authenticated the geometric charts. A dictionary of integer words with the same occurrence matrix is insufficient. The whole-cut wrapper supplies the missing authentication. A `COMPLETE` result from that wrapper means an exact census of handlebodies in the *supplied cut*. It carries no standalone knot verdict. In particular, one ball component in a multi-component cut does not establish that the original exterior is a solid torus.

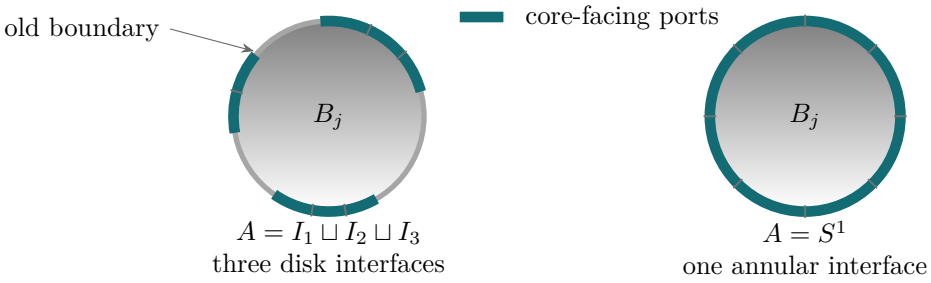


Figure 1: Schematic base boundaries. Subdividing a blue arc into source ports does not subdivide its attaching disk. If every boundary arc faces the core, the complete vertical interface is an annulus. For a disk base, the entire right-hand bundle is a two-handle, as proved later.

3 Ordered parallelity interfaces with polynomial encoded cost

The source parallelity construction already retains the complete base surfaces of the selected product handles, their fibre twisting, and a bounded collection of source-labelled arcs where they meet the complementary core. Its component profiles record which ports belong to a base, but that set does not specify the cyclic order of the ports on its boundary. In particular, several successive port rectangles can form one attaching disk. Counting them as independent attaching disks changes the topology of the proposed gluing.

This section supplies an exact ordered interface operator. It reconstructs the ported boundary circles, distinguishes a common endpoint from a nonempty original-boundary gap, transports the local fibre charts, and groups the ports into the actual connected attaching

intervals and circles. All of this uses binary rank data. The construction does not walk through the potentially enormous collection of individual normal polygons.

The relevant predecessor is the source geometry in the repository's `parallelity.tex`, together with its independently reconstructed cut-cell reference [6]. The orbit-counting engine is the Agol–Hass–Thurston algorithm, abbreviated AHT [1]. The results below separate three points: the already established polynomial weighted AHT operation, a sharper bound for sparse marked weights, and the new extraction of ordered topological attachments. No priority claim for sparse weighted orbit computation is intended.

3.1 The encoded marked-boundary problem

Definition 3.1 (Marked polygonal interval bundle)**.** An encoded marked polygonal interval bundle consists of a compact, possibly disconnected surface B, an interval bundle $E \to B$, and a polygon-family description with the following properties.

1. Polygon copies are indexed by finite integer intervals whose endpoints are written in binary. Each family has an explicitly listed finite sequence of sides and corners.
2. Glued side families have partial affine rank maps of slope $+1$ or -1, an explicit permutation of their two endpoints, and a separate bit specifying whether the interval fibre reverses. These data describe genuine surface and bundle gluings.
3. Exactly P distinguished free edges are individually named by the integers $0, \dots, P-1$. Each occurs at one specified rank and has a local direction from endpoint zero to endpoint one. Their union is denoted A. Every other free edge has the single label "original boundary."

The original-boundary label in this definition carries no additional marked points, curves, slopes, or subdivisions that must be preserved.

In the normal-cut application, the distinguished edges are the source core ports. The source constructor accepts its existing validated finite, connected, compact, orientable triangulation with one torus boundary and an admissible integral standard normal vector. It allows empty, disconnected, and one-sided cutting surfaces. Its geometry has at most $66T$ ports, where T is the number of tetrahedra. If D is the number of normal disks, its corner universe has at most $32D$ points and is represented by $O(T)$ families and side records [6].

The separate fibre bit is essential. An order-reversing map of a one-point rank interval has the same point map as an order-preserving one. Neither that map nor the endpoint permutation alone specifies fibre transport. The new construction retains all three pieces of information.

The output concerns precisely those boundary circles of B that meet A. It does not list every unported boundary circle. Their number and the base topology remain available from the existing base-profile construction.

3.2 Dense weighted transport and the sharper sparse invariant

We first make the complexity accounting explicit. This also corrects an overly cautious reading of later repository passages that express their cost only in terms of realized weighted partitions. The preceding repository section `suffix_folds.tex` already gives a polynomial bound for those partitions. Classical weighted AHT is likewise polynomial [1, Theorem 16]. There is no missing general dense-weight theorem at this stage of the source-cut calculation.

Lemma 3.2 (The existing suffix-fold bound)**.** *Let a valid forward AHT trace on $[0, N)$ start with at most k pairings and R_0 canonical constant weight runs. Weights may be signed integer vectors. After τ truncations the maintained suffix-only transport has at most*

$$R_0 + 2\tau$$

weight runs. If the trace has E events, the total number of emitted histogram blocks is at most

$$E(R_0 + 2E + 4k + 2).$$

For additive source intervals $[l_j, h_j)$ with weights v_j, every intermediate coordinate has absolute value at most

$$M = \sum_j (h_j - l_j)\|v_j\|_\infty.$$

Proof. This is the suffix-specific argument already established in the repository [6]; we recall it to specify exactly which native operation is being bounded. Suppose a translation truncation retains $[0, C)$ and uses period p, with $b = C - p$. Each removed point is sent to

$$\rho(x) = b + ((x - b) \bmod p) \in [b, C).$$

The retained prefix weights stay in place. Let J_- and J_+ be the old interior changes strictly below and strictly above C, respectively. Extend the removed suffix weight by zero. Its discrete derivative is supported at C, N, and J_+. Summing over residues transports these changes into one period. Consequently every new interior change lies in

$$J_- \cup \rho(J_+) \cup \{b, \rho(N)\}.$$

The change at C, if there was one, disappears. Thus at most two new boundaries replace the vanished suffix boundaries, giving an increase of at most two runs.

For a reversing truncation, the removed suffix is reflected into one retained interval. Its old interior changes are reflected individually; only the two endpoints of the receiving interval can be additional changes. The same bound follows. Deletion, trimming, merger, and transmission leave the point weights unchanged. A static contraction retains an ordered subsequence of the old values and cannot increase the number of runs. Induction proves the first assertion.

No trace operation increases the number of pairings. The complement of their domains and ranges has at most $2k + 1$ static gaps. Intersecting these gaps with a weight partition having R runs produces at most $R + 4k + 2$ pieces, a deliberately conservative bound. Summing over at most E contractions and using $R \le R_0 + 2E$ gives the displayed output bound.

Finally, every original point contribution moves to exactly one current representative or one already emitted orbit. Taking absolute values before adding the contributions proves the mass bound, including signed cancellation. Thus the growth in numerical values costs only logarithmically many additional bits. None of the quotient computations iterates over the number of represented points. □

The useful strengthening for ordered interfaces is independent of the number of truncations.

Theorem 3.3 (Sparse additive orbit transport)**.** *Supply an interval pseudogroup on $[0, N)$ by k partial integral isometries, and supply K singleton weight atoms*

$$(x_j, w_j), \qquad w_j \in \mathbb{Z}^d.$$

Locations may repeat and weights may have either sign. Their weighted orbit histogram is computable in time polynomial in

$$k, \quad \log(N + 2), \quad K, \quad d,$$

and the input weight bit lengths. Along a valid AHT trace, the active weight function is supported at at most K points and has at most $2K + 1$ canonical runs. There are at most K distinct nonzero output weight vectors, with at most one further zero-weight row.

If the atoms have the unique scalar weights $w_j = 2^j$, every active nonzero value encodes a disjoint subset of the K labels. Every nonzero run is a singleton, every nonzero histogram row has orbit multiplicity one, and all mask values have at most K bits.

Proof. Temporarily distinguish the atoms even when their numerical weights are equal. Maintain an allocation of the atoms that have not yet been emitted to current universe points. The weight at a point is the sum of its allocated atoms.

Pairing deletion, trimming, periodic merger, and transmission do not move the allocated atoms. A translation truncation maps each atom in the deleted suffix to its unique retained representative modulo the carrier period. A reflection truncation maps each such atom to its unique reflected representative. When several atoms arrive at the same point, their weights are added. A static contraction emits all atoms at a deleted point as the weight of that completed orbit, then reindexes the remaining points monotonically. No operation creates a second copy of an atom.

Induction proves that the atom subsets at all live points and all completed orbits are pairwise disjoint. Hence at most K live points can have nonzero weight. A function supported on K integer points has at most K nonzero singleton pieces and at most $K + 1$ intervening zero pieces; coalescing equal adjacent values can only improve this count. Every nonzero final orbit contains at least one atom, so there are at most K such orbits and at most K nonzero histogram rows. Arbitrarily many unmarked or cancelling orbits share the single zero row.

Each weight coordinate is bounded in absolute value by $\sum_j |(w_j)_i|$. Its bit length is therefore at most the maximum input coordinate bit length plus $\lceil \log_2(K + 1) \rceil$ and a constant. The maintained quotient/remainder and reflection operations process the bounded run list with binary arithmetic. A trace of E events consequently has polynomial weighted replay cost in the displayed parameters and E. For example, a conservative implementation-level bound is

$$O\big(E(k + K)\log(k + K + 2) + K\log(K + 2)\big)$$

arithmetic operations, with the vector dimension and actual integer operand lengths included in the bit cost. The polynomial classical AHT trace bound then proves the claimed polynomial algorithm [1].

For the unique scalar labels, sums of disjoint subsets of distinct powers of two have no binary carries. Distinct occupied points cannot carry the same nonzero mask, because that would allocate one label to two points. This also rules out a nonzero constant run of length greater than one and rules out two distinct output orbits with the same nonzero mask. The last assertions follow. □

Thus the new bound can be recorded together with the existing one as

$$R \le \min\{R_0 + 2\tau,\, 2K + 1\}.$$

Its advantage here is that K is the number of distinguished endpoints, which is bounded by the source size. It is not the number of normal disks.

3.3 The label partitions in weighted selectors are polynomial too

The existing complete base construction uses weighted least representatives to mark individual quotient vertices and boundary components. Its forward implementation retains intervals of

surviving original labels; its independent checker restores deleted gaps backwards. Both admit an explicit bound that can be combined with Lemma 3.2.

Proposition 3.4 (Tagged selector size)**.** *Consider a complete monotone forward trace with E events, at most k pairings, and R_0 initial weight runs. Let G be the total number of contracted static-gap intervals. Then*

$$G \le (2k+1)E,$$

and there are at most $1+G$ live original-label intervals at any time. The forward tagged selector emits at most

$$E\big(R_0 + 2E + 1 + (2k+1)E + 4k + 2\big)$$

raw tagged pieces. Its coalesced output and its independent backward verification have polynomial size and bit cost in the same parameters.

Proof. Initially the live original labels form one interval, or none when the universe is empty. Removing a contiguous interval of current ranks can split one live original-label interval into two. If instead it crosses several live intervals, it deletes middle intervals and cuts only the two end intervals, with no larger increase in their number. Thus one removed gap increases the live interval count by at most one. Retaining a prefix cannot increase it. This proves the $1+G$ bound. Each contraction has at most $2k+1$ gaps, proving the bound on G.

At a contraction the emission routine intersects three ordered partitions: the constant weight runs, the live source-label intervals in current-rank coordinates, and the static gaps. If their sizes are R, L, and r, the common refinement contains at most $R+L+2r$ relevant pieces. Substitute

$$R \le R_0 + 2E, \qquad L \le 1 + (2k+1)E, \qquad r \le 2k+1$$

and sum over at most E contractions. This gives the stated conservative bound. Sorting and coalescing these explicitly bounded pieces is polynomial, and their endpoints remain binary source labels.

For the independent reverse construction, the selected tagged intervals remain disjoint. Inserting a removed gap has one insertion anchor, which can split at most one currently selected interval. Across all reverse insertions there are at most G such new splits beyond the pieces originally recorded at contractions. Truncations insert no selected points: a deleted suffix contains no least original representative. Binary gap insertion, sorting, and coalescing therefore also have polynomial cost. This argument applies directly to the producer's live-label operations and the verifier's separate backward reconstruction [6]. □

Corollary 3.5 (Polynomial source-parallelity accounting)**.** *For the validated normal-cut source class with T tetrahedra and D normal disks, the source-parallelity base construction, its tagged selectors, and the ordered interface construction below have a polynomial-time realization in T and $\log(D+2)$. No extra hypothesis about an uncontrolled number of weighted or label partitions is required.*

Proof. The source geometry has $O(T)$ records and binary rank endpoints. Its corner and polygon universes have sizes bounded by constant multiples of D [6]. The first two source-parallelity queries are the weighted corner and boundary selectors. Proposition 3.4 bounds their outputs polynomially in their AHT traces. The subsequent mark-placement operation intersects their ordered representative intervals with the $O(T)$ polygon-carrier intervals. A common refinement of two ordered interval families has polynomial size. Hence the weights supplied to the base-component and fibre-cover histograms have polynomially many input intervals. Lemma 3.2 bounds both histograms and their integer sizes. Finally, Theorem 3.3

applies to the two ordered interface queries. Choosing classical polynomial AHT schedules for these finitely many queries proves the corollary. □

The new ordered-interface implementation explicitly selects the original AHT periodic-overlap threshold with `periodic_rule='aht'` in both queries. The pre-existing source-parallelity implementation uses its documented forward AHT engine, including its already justified stronger Fine–Wilf merger. The partition bounds above apply to either valid forward schedule; their conversion to a worst-case input bound invokes the polynomial trace bound for the chosen schedule. In particular, the existing phrases "polynomial in the realized partitions" need not be interpreted as an additional unresolved source-parallelity hypothesis. This is a clarification of local encoded operations. It gives no bound on the number of cutting surfaces or hierarchy states that a complete recognizer must discover.

3.4 Two marked orbit queries recover the boundary words

Give every individual polygon corner its place in a single integer universe $[0, N)$. A gluing of polygon sides identifies their endpoint corners. The source rank maps and corner offsets express those identifications as partial integer isometries. Let V denote the resulting interval equivalence system. Its orbits are the actual quotient vertices of the polygonal surface.

For port p, introduce the two endpoint tokens

$$e = 2p, \qquad e = 2p + 1.$$

Tokens are distinct names even when their corner points are identified or are literally the same source corner. Place the singleton scalar weight 2^e at the corresponding corner. The first query is the weighted orbit histogram of V. It records which distinguished endpoints are the same surface vertex. At a boundary vertex at most two port endpoints can occur, because the boundary is a one-manifold. Accordingly, each nonzero mask has one or two bits and occurs once.

Next add to V the endpoint identifications across every original-boundary free edge, omitting every core-port edge. Call the resulting system G. It records connectivity along the boundary after the interiors of the port edges have been removed. Its marked components are intervals, including the possible one-vertex interval between consecutive port edges. Each has exactly two endpoint tokens.

We do not need to query G separately. Instead, construct its two-sheeted fibre cover $\widetilde{G}$ on $[0, 2N)$. Across an original corner gluing, the source fibre bit either preserves or exchanges the sheets. Across an original-boundary free edge, the local polygon fibre chart preserves them. The lifted token (e, s) receives weight

$$2^{2e+s}, \qquad s \in \{0, 1\}.$$

The second query is the weighted histogram of $\widetilde{G}$. Every marked interval has two disjoint lifts, so the nonzero masks give a perfect matching of the $4P$ lifted tokens. If

$$(e, 0) \longleftrightarrow (f, b),$$

the matching identifies the gap endpoints e, f and its fibre reversal b. The other lift must be

$$(e, 1) \longleftrightarrow (f, 1 - b).$$

Both relations are checked; in particular $e \neq f$.

Theorem 3.6 (Exact ordered interface extraction)**.** *For an encoded marked polygonal interval bundle as in Definition 3.1, two sparse weighted orbit queries determine, for every boundary circle meeting A:*

1. *the cyclic order and local directions of all its port edges;*
2. *whether each intervening original-boundary gap is nonempty or consists of a common vertex;*
3. *the fibre reversal along each gap in the specified endpoint charts;*
4. *the connected components of A, classified as intervals or whole boundary circles, with the endpoint charts and fibre transport of every interval component.*

The output has $O(P)$ records. The queries have $2P$ and $4P$ singleton tokens, respectively, and their weight partitions have at most $4P+1$ and $8P+1$ runs. The construction has polynomial encoded cost. If the total interval bundle is orientable, the fibre parity around every output boundary circle is zero.

Proof. The first query computes actual corner equivalence and therefore tells exactly which endpoint tokens are the same surface vertex. Distinct token names give disjoint bits, so their masks lose no endpoint information. Unmarked vertices contribute only zero weight, regardless of their number.

Consider a boundary circle that contains at least one port. Removing the interiors of its port edges leaves a finite collection of compact intervals. An interval can be a single vertex when two port edges are successive with no intervening original-boundary edge. Its two boundary incidences are the two endpoint tokens that it carries. The system G gives exactly these interval components: its corner relations identify surface vertices, and its additional relations cross precisely the non-port boundary edges. Unported boundary circles and interior vertices carry no endpoint labels.

An interval bundle over an interval is trivial. In the two-sheet fibre cover its two endpoint lifts therefore form two disjoint pairs. The second histogram recovers these pairs and the relative fibre chart. This proves the asserted endpoint matching and fibre reversal. A gap is a single vertex if and only if its two tokens belong to the same V-orbit. Indeed, the forward implication is immediate. For the converse, a positive original-boundary path beginning and ending at that vertex, in addition to the two incident port-edge ends, would give more than two boundary incidences there. That contradicts the boundary one-manifold condition.

Join the recovered gap matching to the fixed port matching $2p \leftrightarrow 2p+1$. Each endpoint is incident to one gap and one port, so these alternating components are exactly the ported boundary circles. Start each component at its least port, entering that port through its endpoint zero. Crossing a port in direction zero to one writes $p+1$; crossing it in the other direction writes $-(p+1)$. Crossing the following gap determines the next port endpoint. This produces a deterministic signed cyclic word, up to the normalization just specified. Every port occurs exactly once in one such word, and there are exactly P gaps between consecutive ports in the aggregate.

Split a cyclic word at every nonempty original-boundary gap. Its remaining maximal runs are precisely the connected interval components of A. If there is no such gap, the entire word is one circle component of A. The local fibre transport along a port is trivial in its own polygon chart. Consequently, the XOR of the intervening gap bits gives the fibre transport of each interval run, and the XOR around a cyclic word gives its fibre monodromy. When the total space E is orientable, its interval orientation character is $w_1(B)$. The restriction of $w_1(B)$ to every boundary circle vanishes, since that circle has an annular collar in B. The total XOR around each boundary word is therefore zero.

Both orbit systems have a number of pairing families bounded by a constant multiple of the input side and gluing records. Doubling sheets doubles the universe and only multiplies the number of families by a constant; it does not expand the universe. Theorem 3.3, applied to $2P$ and $4P$ singleton tokens, gives the stated run and complexity bounds. Decoding masks, forming the two matchings, traversing the P port edges, and splitting the words use polynomial work and $O(P)$ output records. □

Corollary 3.7 (Source-size bound for ordered normal-cut interfaces)**.** *For the source normal-cut geometry, the reduced ordered core interfaces have $O(T)$ records and can be computed in time polynomial in T and $\log(D+2)$, independently of the numerical normal multiplicities.*

Proof. Substitute the source bounds $P \le 66T$, $N \le 32D$, and $O(T)$ polygon-side records into Theorem 3.6. The geometry and the source-normal validation are binary rank operations. If $D = 0$, there are no ports and the empty interface follows directly from the source construction. □

3.5 Attaching intervals, attaching annuli, and the exact invariant

Corollary 3.8 (Connected attachment types)**.** *Suppose the output contains a interval components and c circle components of A. Then*

$$\chi(A) = a, \qquad H_1(A;\mathbb{Z}) \cong \mathbb{Z}^c.$$

In the orientable total-bundle setting, the interval-bundle attachment above each component is a disk for an interval component and an annulus for a circle component. Adjacent source ports in one interval run form one attaching disk.

Proof. A finite disjoint union of compact intervals and circles has the stated Euler characteristic and first homology. The restriction of any interval bundle to an interval is a product, hence a disk. On a boundary circle of an orientable total bundle the fibre character is zero by Theorem 3.6; its vertical restriction is therefore an annulus. The decomposition into components is the maximal-run decomposition already proved, so the several rectangles over consecutive port arcs constitute one connected disk, rather than separate attaching disks. □

The value a agrees with the existing independently checked attachment Euler characteristic, computed there as the number of quotient vertices incident to core ports minus the number of core ports. The new operator retains the extra information that this Euler characteristic omits: the number and identities of entire attaching circles, cyclic order, individual interval membership, and fibre transport.

For example, the four-tetrahedron layered-meridian fixture has 15 ports but only three interval attachments. An especially small annular obstruction uses the two-tetrahedron layered source and the coordinate rows

$$(0,0,1,1,1,0,0), \qquad (1,1,1,1,0,0,0).$$

Its source parallelity base is the edge-cap-only disk already described in the repository's parallelity analysis [6]. The new output has the single word

$$(1,2,3,4,5,6,7,8)$$

and every following gap is a common vertex. Thus $a = 0$, $c = 1$: its whole base boundary is attached, and the corresponding vertical attachment is an annulus. A criterion that permits only attaching disks must exclude this case. It cannot replace the circle by eight independent disk attachments.

3.6 Replication along a two-sided normal surface

The dependence on a large common multiplicity can be separated even more strongly when a global coorientation is available. The following statement concerns the source decomposition itself, including its marked attachment data; it is not merely a statement about the number of complementary components.

Theorem 3.9 (Positive-multiplicity replication)**.** *Let F be a nonempty two-sided properly embedded normal surface with admissible standard coordinate vector x in a triangulated orientable 3-manifold satisfying the source contract. Fix a global coorientation on each component of F. Write K_x for the source complementary core and $E_x \to B_x$ for its selected parallelity bundle. For every integer $m \geq 1$, the standard normal surface with coordinates mx admits the following source-compatible description.*

1. *Its core is canonically identified with K_x by the local handle types and owners. Source rank labels on its interface are obtained by multiplying the corresponding local gap ranks by m.*
2. *In every product-handle family, whose local gap ranks have the form $1 \leq r < md$ after replication, retain exactly the copies with $m \mid r$. These copies form a union of complete bundle components. Division of their local ranks by m identifies their polygonal bundle, including its fibre charts and marked core attachment, with $E_x \to B_x$.*
3. *Every source core port belongs to this divisible-rank subbundle. The remaining product handles form $m-1$ unported copies of $F \times I$, with $m-1$ base copies of F. If F has q connected components, this means $(m-1)q$ additional connected unported bases. Existing unported components of B_x are retained as well.*
4. *The reduced signed core-port words, the zero-or-positive original-boundary gap indicators, and the fibre transport are unchanged under the source identification. In particular, the numbers and owner incidences of attaching intervals and attaching circles are unchanged.*

The identification is by source family and local rank; the flattened integer offsets and the numerical polygon-family IDs need not agree. Topologically,

$$\mathrm{Cut}(M, mF) \cong \mathrm{Cut}(M, F) \sqcup \bigsqcup_{j=1}^{m-1} (F \times I).$$

Proof. Choose a small two-sided normal collar of F compatible with the fixed coorientation. Replace each local disk of F by m parallel disks in that collar. Face matching follows the old matching and the collar coordinate, so this is an embedded normal representative of mx. There are $m-1$ intervening product slabs over every component of F. The remaining part of the complement is the old cut complement, with its two cut sides shifted to the two exterior copies of the collar. This proves the displayed topological decomposition. We now verify its stronger assertion about the specified source handles.

A local product handle lies between consecutive objects in one ordered stack: parallel normal disks in a tetrahedron, parallel normal arcs in a face, or normal intersection points on an interior edge. If the old stack has size d, its product ranks are $1, \ldots, d-1$. Replication replaces each old object by a block of m objects. The gap between successive old blocks has rank mr, whereas an interior gap within a block has rank not divisible by m. Thus the divisible ranks are exactly the old inter-block product handles. If an old stack has $d = 1$, its source family was absent; the new ranks $1, \ldots, m-1$ are all nondivisible, as required.

This distinction is respected by every source gluing. Expressed in *local gap ranks*, each gluing is $r \mapsto \varepsilon r + c(x)$, where $\varepsilon \in \{1, -1\}$ and $c(x)$ is a homogeneous integral linear combination

of the relevant normal disk counts. For example, the prism-to-face maps are r, $t_v + r$, or $t_v + q - r$, where t_v is a triangle count and q the relevant quadrilateral count. The prism-to-edge and face-to-edge maps have the forms $t_v + r$, $w - t_v - r$, r, or $w - r$, where w is an edge weight; restrictions choose the appropriate family and interval. These are gap ranks, so the -1 that occurs in a zero-based *disk* reflection has already been removed by the source conversion to gap coordinates. After replacing x by mx, the offset becomes $mc(x)$. Hence divisible ranks glue only to divisible ranks, and division by m recovers every original gluing and its domain. The endpoint permutations and fibre-reversal bits depend on source orientations and local disk types, which have not changed. This proves the asserted bundle identification, including local fibre charts.

The complementary source 0-handles are the chambers at stack ends and at changes of disk type. Its 1-handles are the corresponding nonparallel face regions, its 2-handles are the two exterior edge gaps (or the sole gap for zero edge weight), and its 3-handles are the fixed interior vertex regions. Positive scaling preserves all zero-or-positive tests and disk-type supports. Consequently their types, owners, and attaching charts identify with those for x. Each core-facing product port occurs at a transition rank such as t_v, $t_v + q$, an arc count, or its complement in an edge weight. These ranks are homogeneous in x, so all new port ranks are divisible by m and all old ports reappear at their rescaled ranks. The possibly newly present product families with old stack size one have no such interior divisible rank and contribute no core port.

The product handles at the nondivisible ranks fill precisely the slabs inside the replicated collars. The chosen global coorientation orders the m sheets consistently on each component of F, thereby identifying the slabs as $F \times I$ with indices $1, \dots, m-1$. A reversed local stack can exchange residue s with $m - s$; this local effect does not exchange the globally indexed slabs after the coorientation is used. Their bases are the corresponding copies of F and their only vertical boundary lies over ∂F in the original boundary of M. None touches the source core.

Finally, the divisible-rank identification preserves all free-edge labels, corner identifications, and fibre bits. It therefore preserves the two pairing problems used by Theorem 3.6 on every ported component. The reduced boundary words, their gap types, and their attaching-component decomposition are identical after applying the source identification. This proves all four assertions. □

Corollary 3.10 (Replication of a certified disk cut)**.** *If F is a connected two-sided normal disk and its primitive source cut has a valid handlebody census, then every positive multiple has the same ported core and contributes $m - 1$ additional unported balls. In particular, if the primitive cut is a single ball, the cut along mF consists of exactly m balls.*

Proof. Apply Theorem 3.9 and use $D^2 \times I \cong D^3$ for the additional unported slabs. □

This theorem permits a future certificate format to retain a primitive interface proof and a binary replication factor, once a source-bound coorientation certificate has been checked. The delivered implementation still runs its general compressed queries on the supplied vector; it does not use such a replication shortcut. The coorientation hypothesis is substantive: for a one-sided component the local sheet order can reverse around a loop, and the asserted list of $m - 1$ copies of F does not follow. The general ordered-interface operator itself remains valid in that case.

For the layered-meridian experiments below, the theorem proves independence of the positive scalar m once the primitive disk and its source interface have been certified. It does not prove an interface formula for all numbers of layers. In particular, the observed formulas for the number of primitive ports and attaching intervals as the layer count varies remain

finite experimental observations here; no induction establishing an all-layer primitive-interface theorem is claimed.

3.7 Source-bound certificates and implementation contract

The producer is `normal_parallelity_boundary_words`; the independently implemented checker is `verify_parallelity_boundary_words`. The result retains the source port records, including their core-handle owners, and contains the following finite data.

- Each gap specifies its two endpoint token IDs, whether original boundary is present, and its fibre reversal. Endpoints are ordered by ID.
- Each ported boundary circle contains its signed word, the corresponding list of following gap IDs, and the whole-core-circle indicator.
- Each connected attachment contains its interval-or-circle type, its signed subword, and, for an interval, its entry and exit tokens and fibre transport. A least port ID serves as an anchor for joining it to the existing base profile whose port mask contains that ID.

All ports in one connected attachment belong to the same base component. A base profile containing a port has multiplicity one in the existing source classification. Thus the anchor joins the ordered interface to the base topology without making any choice among repeated unported components.

The certificate has schema `parallelity-boundary-words-v1`. It binds the input fingerprint, the proposed interface, the weighted vertex-partition proof, and the weighted gap-cover proof. Verification reconstructs the parallelity geometry through the separate raw-cut-derived reference. It independently forms the two pairing systems, checks the supplied AHT traces and weights, and reconstructs the proposed words and connected attachments. It does not import the producer's query builder, mask decoder, traversal routine, or discovery scheduler.

When $P = 0$, the output is the empty interface. Both proof fields are explicitly null; neither orbit query is run. The checker reconstructs the source and verifies the absence of ports, so this shortcut does not rely on a caller assertion. It says nothing about the number of unported base boundary circles, which this API has not been asked to enumerate.

A discovery-cycle limit is shared by the two queries. Exhaustion returns an inconclusive status without an interface or certificate. A verification operation limit counts the two supplied ordinary traces. Cooperative cancellation propagates through both paths. The operation returns geometric information about the supplied source and cutting vector; it does not itself issue a knot verdict.

3.8 Independent finite checks and binary scaling

The added test module passed all eight test methods in the recorded run. It covers 193 source/vector cases: all 157 saved native source cases, 16 layered-meridian cases, eight one-sided cases, and twelve source relabellings. Of these, 182 are compared with a literal expanded oracle. That oracle expands the actual free polygon edges, identifies corner fibre charts with a parity union-find, and walks each old-boundary edge one at a time. It separately computes connected components using only the core edges, rather than reusing the producer's maximal-word-run routine. Eleven very large native cases use compressed independent source replay and the attachment-Euler and base-profile checks instead of exceeding the literal oracle's point cap.

Further tests reject altered source fingerprints, core owners, signed words, zero-gap flags, fibre bits, endpoint masks, and attachment types. They verify exact hexadecimal JSON replay

with multiplicity 2^{16384}, shared caps, cancellation, and producer-disabled verification. Four generic sparse-token examples compare weighted translation orbits with their exact residue classes, up to a 16384-bit universe. The test log is `boundary-words-tests.txt`; the recorded run took 69.940 seconds.

The separate benchmark uses actual layered-solid-torus normal coordinates with $T \in \{1, 2, 4, 8\}$ and

$$m = 2^e, \qquad e \in \{0, 16, 64, 256, 1024, 4096, 16384\}.$$

All 28 certificates are independently replayed. Selected measurements are shown in Table 2. They measure the new ordered-interface operator, not a complete unknot-recognition run. They are individual scale measurements, not statistical or competitive timing claims.

T	m	P	a	Produce (s)	Verify (s)	Certificate bytes
2	1	2	2	0.00342	0.00371	5,949
2	2^{16384}	2	2	0.17363	0.09245	4,541,499
4	1	15	3	0.10266	0.08041	47,030
4	2^{16384}	15	3	1.12275	0.65597	12,861,283
8	1	47	3	0.70518	0.53165	164,528
8	2^{16384}	47	3	4.47206	2.50908	30,982,516

Table 2: Exact source-bound ordered interfaces. Here a is the number of connected interval attachments, not the number P of source ports.

At $T = 2$ and $m = 2^{16384}$, the largest vertex and cover weight partitions have 9 and 17 runs, meeting the respective sparse upper bounds. At $T = 4$ they have 49 and 97 runs, and at $T = 8$ they have 145 and 289. These remain below $4P + 1$ and $8P + 1$. The one-tetrahedron examples have no ports and use the no-query shortcut; their boundary-interface certificate has 294 bytes at every measured exponent.

The distinction between output structure and proof size is visible in the table. The number of words, ports, and connected attachments stays fixed for each source family. Binary endpoints in the local-reduction traces still grow with the multiplicity bit length. In particular, the eight-tetrahedron extreme example has a roughly 31-million-byte certificate, despite its 47 ports. The data support polynomial encoded scaling and constant combinatorial interface size in these families; they do not support a claim of constant certificate size.

3.9 An alternative certificate using only three orbit counts

The following elementary result supplies an alternative way to authenticate the marked partitions. It is useful both as a separate proof principle and as a possible future checker. It is not an implemented backend of the present package.

Theorem 3.11 (Three-count certificate for a marked partition)**.** *Let an interval equivalence system have C orbits and let $K > 0$ marked tokens be placed at specified points. A proposed partition of the tokens into m nonempty blocks is exactly their original orbit partition if and only if the following two tests hold.*

1. *Add singleton pairings forming a spanning tree within each proposed block. The resulting orbit count is still C.*
2. *In a separate copy of the original system, choose one representative token from each proposed block and add singleton pairings forming a tree on these m representatives. The resulting orbit count is $C - m + 1$.*

Thus the original count and these two augmented counts form a three-trace certificate of the marked partition. The $K = 0$ case is immediate and requires no partition query.

Proof. Adding a pairing can only merge original orbits; it cannot split one. Therefore the first augmented count equals C precisely when every edge of every within-block tree has its ends in the same original orbit. By connectivity of a tree, all tokens in each proposed block then belong to one original orbit.

Let q be the number of distinct original orbits occupied by the chosen representatives. Connecting all representatives into one tree merges exactly those q orbits into one, irrespective of repetitions among the representatives' original orbits. The second augmented count is therefore $C - q + 1$. Requiring it to be $C - m + 1$ is equivalent to $q = m$, so the proposed blocks belong to distinct original orbits. Together with the first test, this proves exactness. Conversely, an exact partition plainly satisfies both count identities. □

There are at most $K - 1$ added singleton pairings in either augmentation. For discovery, equality of two marked points can likewise be tested by adding one pairing and checking whether the orbit count drops. Performing at most $\binom{K}{2}$ such tests constructs the entire partition in polynomial time from the unweighted AHT oracle. For sparse integer weights, summing the weights over these marked blocks then recovers their histogram; the remaining unmarked orbits, together with any zero-sum marked blocks, supply the zero row. This gives a slower but conceptually independent polynomial alternative to Theorem 3.3. The shipped two-query construction uses the more economical sparse weighted transport and its separate local replay checker.

3.10 What the ordered interface does and does not settle

Replacing each nonempty unmarked boundary path by one recorded gap preserves the marked boundary pair and the relative fibre charts needed for the interval-attachment calculation. If a later hierarchy has additional pattern marks inside those paths, those marks must be distinguished before compression, with their encoded number included in the input bound. The present one-color original-boundary contract cannot silently preserve such extra data.

The construction also does not classify arbitrary annular attachments, make the inherited core admissible, produce a new compressing disk, or bound the number of hierarchy restarts. Its concrete advance is a source-bound, polynomial extraction of the exact connected and ordered attachments that the next topological gluing operation requires. Together with the dense and tagged-selector bounds above, it establishes local encoded cost without turning that local statement into a global quasi-polynomial recognition claim.

4 Geometric certificates for the core and its marked attachments

A bounded complementary core is useful only if its topology can be certified and the omitted product regions can be glued back correctly. This section supplies a conservative consumer of the existing native normal-cut geometry. The consumer has two parts. First, it certifies handlebodies by cancellations witnessed by *actual geometric intersection counts*. Second, it reconstructs the topology of an assembly whose interfaces are separated disks. A complete polynomial criterion is obtained for the ball cores accessible to raw singleton cancellation. An explicit infinite family shows that this criterion can succeed uniformly when the normal coordinates grow exponentially.

The algebraic source-closure principle used below is already present in ProveIt report 56, `reports/56/article/raw_epoch.tex` [6]. We prove the needed matrix statement for completeness. The contribution here is its geometric realization and its application to source-

authenticated normal-cut handles. Neither the classical handle cancellation theorem nor the elementary topology of interval bundles is claimed as new.

4.1 The geometric input and the role of literal occurrences

We use the full geometric conclusion of the existing source-parallelity construction, described in `synthesis/parallelity.tex` [6]. This conclusion supplies an embedded interval bundle, its complete base polygons and fiber identifications, and an inherited complementary handle structure. The polygon construction includes contacts of dimensions zero–one, zero–two, and one–two, together with the source edge and vertex strata. Its core records contain both ends of every one-handle, the alternating attaching sectors of every two-handle, and the vertex-link charts of every three-handle. We invoke this geometric conclusion, rather than infer an embedding from Euler characteristics, component histograms, or an abstract presentation.

Definition 4.1 (Authenticated core incidence data)**.** Let K be a compact orientable three-manifold with a genuine finite handle decomposition. Suppose that it has no three-handles. Its zero- and one-handles form a disjoint union of handlebodies. Contract a spanning forest of their incidence graph by geometric zero/one-handle cancellations. In one resulting connected component, let

$$D_1, \dots, D_k$$

be the complete system of co-core disks of the remaining one-handles, and let $\alpha_1, \dots, \alpha_f$ be the disjoint embedded two-handle attaching curves on the boundary of this handlebody. Put the curves and disk boundaries in the transverse position specified by the handle charts and set

$$n_{ih} = \left|\alpha_i \cap \partial D_h\right|. \tag{G1}$$

The matrix (n_{ih}) is authenticated when every counted occurrence is an actual passage through the corresponding source one-handle, with its source attaching sector retained.

A forest cancellation only absorbs tree one-handles into zero-handles. The other co-core disks survive, so deleting the tree-handle entries from the source attaching lists gives the incidences in (G1). A two-handle cannot connect distinct components of the zero/one-handle union: its attaching circle is connected. Thus the forest also identifies the components relevant to all subsequent cancellations.

This contract is stronger than a cellular boundary matrix or a fundamental-group presentation. An exponent sum of 1, or a letter occurring once after free reduction, need not mean that an attaching curve meets a belt circle once. All occurrences in (G1) are unsigned *geometric* occurrences. There is no free reduction, cancellation of opposite signs, or replacement by a homotopic curve hidden in this definition.

4.2 A geometric realization of positive incidence elimination

Lemma 4.2 (Meridian-disk realization of a raw singleton pivot)**.** *Let H be an orientable handlebody with a complete disk system $D_1, \dots, D_k$, and let $\alpha_1, \dots, \alpha_f$ be pairwise disjoint embedded attaching curves on ∂H. Write $n_{ih} = |\alpha_i \cap \partial D_h|$. Suppose that $n_{rg} = 1$. Attach the two-handle with attaching curve α_r. The resulting manifold is a handlebody of genus $k-1$ with a complete disk system indexed by $h \neq g$ for which the surviving attaching curves have incidences*

$$n'_{ih} = n_{ih} + n_{ig} n_{rh}, \qquad i \neq r, \quad h \neq g. \tag{G2}$$

In particular, the equality counts actual intersections; it is not merely an equality of homology classes or group words.

Proof. Keep D_g fixed, and write q for its unique intersection with α_r. If α_r meets another disk boundary, choose such an intersection p adjacent to q along α_r, among all current intersections with disk boundaries. Suppose that p lies on ∂D_h. The intervening arc has interior disjoint from all disk boundaries. It is also disjoint from every other attaching curve, since those curves are disjoint from α_r.

Slide D_h over a parallel copy of D_g along a thin band following this arc. Choose the copy on the side of D_g approached by the arc, and choose its endpoint just before q. The band then misses D_g itself. The standard boundary connected sum replaces D_h by a properly embedded disk and preserves a disjoint complete disk system. This elementary disk slide can be seen in a regular neighborhood of the two disks and the band: sliding one disk across the parallel copy changes the chosen meridian system without changing the handlebody cut out by it.

The smoothing at the band removes the intersection at p and the unique intersection of α_r with the parallel copy of ∂D_g. The two sides of the band miss α_r. Thus the replacement for D_h has exactly one fewer intersection with α_r; the intersections of α_r with the other disks are unchanged. Small band endpoints and a sufficiently thin band miss all other attaching curves. For any $i \neq r$, the new disk boundary consequently consists, for purposes of counting intersections with α_i, of the old ∂D_h and one parallel copy of ∂D_g. Its intersection count increases by exactly n_{ig}.

Repeat this operation, always using the fixed disk D_g and an adjacent intersection along α_r. At each step the total number of intersections of α_r with the other disk boundaries decreases by one. Hence the process terminates. The disk indexed by h has been slid exactly n_{rh} times, whereas D_g and its intersections with the other attaching curves have remained fixed. The resulting disks E_h, $h \neq g$, are disjoint from α_r and satisfy

$$|\alpha_i \cap \partial E_h| = n_{ih} + n_{ig}n_{rh}.$$

In the disk system consisting of D_g and the E_h, the selected attaching curve goes over the handle dual to D_g exactly once and misses all other co-cores. Its one/two-handle pair therefore cancels by the geometric one-point cancellation theorem; see, for example, [5, Section 10.1.3, Lemma 10.3]. Equivalently, cutting along the E_h leaves a solid torus, and attaching a two-handle along a curve meeting its meridian once turns that solid torus into a ball. Thus the E_h form a complete disk system after attachment. Every other attaching curve is disjoint from α_r, so it lies on the new boundary, and its intersections with the E_h are unchanged. This proves (G2). □

Remark 4.3 (What the construction does and does not record)**.** The proof permits very many disk slides. Its purpose is to prove that a specific integer update is geometrically realizable. A classifier can verify that update without expanding the disks or constructing the resulting boundary homeomorphism. If a later hierarchy operation needs an explicit lifted disk or the transported position of a peripheral curve, that additional output must have its own representation and complexity bound. The present certificate asserts the handlebody type and the existence of the indicated complete disk system.

Proposition 4.4 (Saturation removes occurrence growth from verification)**.** *Let $C = \{0, 1, 2\}$ have operations*

$$a \oplus b = \min\{2, a + b\}, \qquad a \otimes b = \min\{2, ab\},$$

and let $\sigma(n) = \min\{2, n\}$ for $n \in \mathbb{N}$. Starting from authenticated incidences, store only $c_{ih} = \sigma(n_{ih})$. A pivot (r, g) is geometrically valid whenever $c_{rg} = 1$. After deleting its row and column, the exact saturated incidences are

$$c'_{ih} = c_{ih} \oplus (c_{ig} \otimes c_{rh}) = \min\{2, c_{ih} + c_{ig}c_{rh}\}. \tag{G3}$$

For $k+f = O(T)$, a supplied sequence removing all two-handles is replayed in $O(T^3)$ operations on constant-sized entries and $O(T^2)$ working space. Its forest and pivot labels use $O(T \log(T+1))$ bits.

Proof. The map σ preserves zero, one, addition and multiplication when its codomain uses $\oplus$ and $\otimes$. This follows directly by separating the cases 0, 1, and at least 2. In particular, $\sigma(n) = 1$ if and only if $n = 1$. Lemma 4.2 and induction therefore show that every stored matrix is exactly the saturation of an actual geometric incidence matrix. No large integer or word needs to be reconstructed. There are at most $\min\{k, f\}$ pivots, each of which updates at most kf entries. The remaining bounds follow from the matrix size and from recording $O(T)$ source identifiers. □

The operation bound concerns this matrix phase after the source charts have been authenticated. It does not absorb the cost of input validation, normal-coordinate arithmetic, the interval-orbit proofs used to obtain the parallelity data, or a preceding search for the normal surface.

Corollary 4.5 (Core handlebody certificate)**.** *Consider a connected authenticated core with v zero-handles, e one-handles, f two-handles, and no three-handles. If a sequence of valid pivots removes every two-handle after spanning-tree contraction, the core is a handlebody of genus*

$$g = e - v + 1 - f. \tag{G4}$$

Proof. Tree contraction leaves one zero-handle and $k = e - v + 1$ one-handles. Each pivot attaches one of the remaining two-handles and reduces the handlebody genus by one, by Lemma 4.2. After all f attachments have been made, no further handles remain. The result is a handlebody of genus $k - f$. □

The hypothesis excluding three-handles is essential to this conclusion. The actual attaching curves and their disjointness are essential as well: an arbitrary integer matrix is not a certificate of a three-manifold's topology. Failure of a search for these pivots does not show that the core is not a handlebody.

4.3 A complete polynomial criterion within the ball-core class

There is a substantial improvement over trying arbitrary singleton pivots. In the square case, an entire successful raw elimination can be discovered without forming any fill entries. This is the geometric application of the immutable-source theorem from report 56 [6].

Lemma 4.6 (Reversing a successful positive unit pivot)**.** *Let $A = (a_{ij})$ be a square matrix of nonnegative integers with $a_{dc} = 1$. For $i \neq d$ and $j \neq c$, put*

$$b_{ij} = a_{ij} + a_{ic}a_{dj}.$$

If $\operatorname{per}(B) = 1$, *then* $\operatorname{per}(A) = 1$. *The permanent of the empty matrix is understood to be* 1.

Proof. View a matrix entry as a number of separately labelled edges in a bipartite multigraph. A permanent counts its edge-labelled perfect matchings. Tag each edge of B either as a direct edge of A, or as an ordered pair consisting of an edge from i to c and an edge from d to j.

The unique tagged perfect matching of B contains at most one pair-created edge. If it contained two, say in positions (i, j) and (k, ℓ), then $i \neq k$ and $j \neq \ell$. Exchanging their destination factors would produce pair-created edges in positions (i, ℓ) and (k, j), giving a

second tagged perfect matching. If the unique matching contains no pair-created edge, adjoin the unique edge (d, c) to obtain a perfect matching of A. If it contains one, replace that edge by its two component edges to obtain a perfect matching of A.

Conversely, map any perfect matching of A into a tagged perfect matching of B. If it uses (d, c), delete that edge. Otherwise there is a unique row matched to c and a unique column matched from d; replace these two edges by the corresponding pair-created edge. This map is injective: the zero-or-one number of pair-created edges distinguishes the cases, and the tags recover all original edge labels. Hence A has at least one and at most one perfect matching. □

Definition 4.7 (Source row-leaf deletion)**.** In a matrix with some rows and columns already deleted, a row is a *source row leaf* if its immutable source entries have exactly one nonzero among the live columns, and that entry equals 1. A row-leaf deletion removes this row and its unique live column. No fill update is performed.

Theorem 4.8 (Complete source criterion for raw ball-core cancellation)**.** *Let a connected authenticated core have no three-handles and satisfy*

$$k = e - v + 1 = f.$$

Let A be its square unsigned incidence matrix after spanning-tree contraction. The following conditions are equivalent:

1. *Some sequence of positive singleton pivots* (G2) *removes every row and column.*
2. *The source matrix satisfies* $\operatorname{per}(A) = 1$.
3. *Source row-leaf deletion removes every row and column.*

Whenever these conditions hold, the core is a three-ball. Condition 3 is decidable in $O(k + f + I)$ structural operations after source preparation, where I is the number of nonzero source incidences. A successful trace has a verifier with the same linear bound and $O((k + f)\log(k + f + 1))$-bit labels.

Proof. A successful sequence in condition 1 ends in the empty matrix. Apply Lemma 4.6 backwards through the sequence to obtain condition 2.

Suppose condition 2 holds. Let its unique edge-labelled perfect matching assign one column to each row. Every matched entry equals 1, since two parallel edges in such an entry would give two matchings. Name each row by its matched column, and draw a directed edge $x \to y$ whenever $y \neq x$ and that row has a nonzero entry in column y. A directed cycle would allow the matched columns along the cycle to be exchanged, producing a second perfect matching. The directed graph is therefore acyclic. A sink row has no nonzero entry except its unit matched entry, and is a source row leaf. Deleting it and its column leaves a matrix whose permanent is still 1. Induction proves condition 3.

Finally, a row leaf has zero in every other live column. Thus the fill term in (G2) vanishes for all surviving entries. Every source row-leaf deletion is consequently a valid positive singleton pivot, and condition 3 implies condition 1. Lemma 4.2 realizes these pivots geometrically. Corollary 4.5 gives genus $k - f = 0$.

For discovery, store each row's live support size and an incidence list from each column to its source rows. Put the eligible rows in a queue. On deleting a column, update its incident rows; each source incidence is processed a bounded number of times. A row enters the queue only when its live support becomes one, with stale entries discarded if its support subsequently becomes zero. This gives the stated structural bound.

An independent checker need not recreate a sequence of mutable matrices. Assign each deleted column its deletion time. For every supplied pivot, check that its source entry equals

1, and that every other source column in that row has an earlier deletion time. Check that each row and column appears exactly once. These source conditions state precisely that each step is a row-leaf deletion. Scanning the immutable incidences verifies all conditions in linear structural work. Identifiers, times and counters have logarithmic bit length. □

Remark 4.9 (Scope of completeness)**.** The theorem is complete for the specified raw geometric cancellation model on the supplied handle structure. It is not a recognition theorem for all ball cores. A ball can require a different disk system, more general slides, a different spanning tree, or other geometric operations before this model succeeds. Even when the source permanent is one, a poor arbitrary singleton pivot can create positive fill and become stuck. The source row-leaf procedure avoids this order dependence. The permanent is used only in the proof and is never computed by the algorithm.

Proposition 4.10 (A bounded-genus extension)**.** *Let a connected authenticated core have k non-tree one-handles, f two-handles, no three-handles, and $g = k - f \geq 0$. Fix a set S of g one-handles to survive. A complete raw singleton cancellation with these survivors exists if and only if the square source matrix on all rows and on columns outside S has permanent one, equivalently if source row-leaf deletion succeeds when the columns in S are treated as already known. Therefore exhaustive search over survivor sets is complete within this raw model in*

$$\binom{k}{g}\,\mathrm{poly}(T) \tag{G5}$$

work when $k + f = O(T)$. This is polynomial for fixed g and $2^{O((\log(T+1))^2)}\,\mathrm{poly}(T)$ when $g = O(\log(T+1))$.

Proof. Restrict any successful positive-pivot sequence to the non-survivor columns. The restricted square matrix follows the same positive-pivot rule and ends empty, so its permanent is one by Lemma 4.6. Conversely, apply the row-leaf order of Theorem 4.8 to this restricted matrix. A donor row can have arbitrary source entries in S. Previously eliminated columns have images supported only in S and still earlier columns. Consequently they introduce no occurrence of a different live non-survivor column. The unique live non-survivor occurrence remains exactly one, and every cancellation is geometrically valid by Lemma 4.2. All $f = k - g$ rows are consumed. Enumerate the $\binom{k}{g}$ possible survivor sets and apply the source test to each. Finally $\binom{k}{g} \leq k^g$ gives the displayed asymptotic consequences. □

The exhaustive survivor search in this proposition is a theoretical extension of the core certificate, not an assertion that the delivered implementation searches every such set. Its geometric use is new to this consumer; the corresponding immutable-source closure principle was already proved in report 56 [6]. It also does not bound a hierarchy's genera or prove that every unknot enters this cancellation class.

4.4 Interval bundles and the actual marked interface

Lemma 4.11 (An orientable interval bundle over a bordered surface)**.** *Let F be a connected compact surface with nonempty boundary, and let $p : P \to F$ be an interval bundle whose total space is orientable. Then P is a handlebody of genus*

$$1 - \chi(F). \tag{G6}$$

If F is orientable of genus h with $b > 0$ boundary components, this genus is $2h + b - 1$. If F is nonorientable with q crosscaps and $b > 0$ boundary components, it is $q + b - 1$.

Proof. A connected compact surface with nonempty boundary admits $1-\chi(F)$ disjoint properly embedded arcs that cut it into a disk. For example, take the co-core arcs in a decomposition of F into one disk and $1-\chi(F)$ surface one-handles. The inverse image of each arc under p is a properly embedded disk. Cutting P along these disks gives the interval bundle over a disk, namely $D^2 \times I$, a three-ball. The disks are therefore a complete meridian system. Orientability of the total space ensures that reconstructing it uses orientable three-dimensional one-handles, proving the handlebody assertion and the genus formula. The two displayed specializations follow from surface Euler characteristic. □

In particular, the orientable interval bundle over a Möbius band is a solid torus. Requiring the base to be orientable would incorrectly discard this useful case. Conversely, the nonempty-boundary hypothesis must be checked: for example, $S^2 \times I$ is not a handlebody, and the arc-cutting argument does not apply to a closed base.

Lemma 4.12 (Maximal marked runs give the disk interfaces)**.** *Assume the full existing source-parallelity conclusion: in the cut three-manifold the selected total interval bundle $p : P \to F$ is embedded, its complementary inherited core is K, and*

$$P \cap K = p^{-1}(A), \qquad A \subseteq \partial F, \tag{G7}$$

where A is the union of the closed core-labelled base arcs, with all source endpoint contacts included. On each completed base boundary circle, amalgamate consecutive core-labelled arcs into maximal connected runs. If no run is a whole boundary circle, then every component of $P \cap K$ is an embedded disk. Distinct such components have separated closures and admit compatible corner collars. If a run is a whole boundary circle, its inverse image is an annulus and is outside this disk-interface conclusion.

Proof. The completed base boundary is a compact one-manifold. A finite union of its closed cell arcs has connected components that are either closed intervals or entire circles; an interval is maximal precisely after all consecutive selected arcs have been amalgamated. Distinct interval components are separated by nonempty open complementary arcs. An interval bundle over a closed interval is a disk. Over a boundary circle of a surface, an interval bundle with orientable total space restricts to an annulus: the base has an annular collar there, so orientation of the total space rules out reversal of the fiber around that circle.

Equation (G7) is a statement about the embedded bundle and core, not merely their quotient data. Thus each inverse-image disk is embedded also on the core side. Distinct components are disjoint compact subsets of this embedded product boundary; there are only finitely many core arcs, and hence finitely many such components. Choose disjoint small neighborhoods and use the fiber charts and boundary collars to round their corners compatibly. The rounded disks still have disjoint closures. An apparent shared corner between different bundle components, or an alternating corner that was not identified in the completed base, would contradict the assumed embedded interval-bundle structure and its full product-intersection formula. This is why the source theorem's edge and vertex contacts are needed. □

The existing geometric theorem supplying (G7) is the trusted mathematical input here; its source charts are reconstructed by the geometry verifier [6]. The new ordered boundary word certifies which maximal components of A are intervals and which are circles. A cyclic word scan alone would not establish (G7). In particular, a single face port is not automatically one disk interface. Several ports can constitute one disk, or together form an annulus. Independently shrinking every face port would change the attachment and is not a valid substitute for using its true maximal run.

Each maximal run has one owner component of the core, as verified from the source attachment charts. This owner is a core *component*, not necessarily one core handle: consecutive arcs can face distinct handles within that same component.

4.5 Disk gluing gives an exact genus formula

Theorem 4.13 (Handlebodies glued along separated disks)**.** *Let $\{H_v : v \in V\}$ be finitely many connected orientable handlebodies of genera g_v. Specify pairs of embedded closed disks on their boundaries, all with pairwise disjoint closures before gluing, and a homeomorphism for each pair. Suppose that the quotient, with corners rounded, is orientable. Let Γ be the incidence multigraph, with one vertex per handlebody and one edge per disk pair; loops and parallel edges are allowed. Every connected component C of Γ gives a handlebody component of the quotient, of genus*

$$G_C = \sum_{v \in V(C)} g_v + |E(C)| - |V(C)| + 1. \tag{G8}$$

It is a three-ball if and only if every constituent genus is zero and C is a tree.

Proof. The pieces meet only in the specified disk identifications, so the graph records connectedness exactly. Work with one connected graph component and choose a spanning tree.

Choose a complete meridian system in each H_v that misses all its marked disks. Such a choice always exists: first move the finitely many marked boundary disks, by a boundary isotopy, into small disjoint neighborhoods away from any chosen meridian boundaries; extend the isotopy over a collar of the handlebody and pull the meridian system back. Thus this step does not impose an extra condition on the placement or gluing maps of the marked disks.

Cut the quotient along all these local meridian disks. Also cut along the gluing disk corresponding to each edge outside the chosen spanning tree. The remaining manifold is a tree of balls glued along boundary disks. It is a ball, by repeatedly deleting a leaf ball and applying the elementary ball-attachment lemma; compare [4, Lemma 1.3]. Consequently the local meridian disks together with the non-tree gluing disks form a complete meridian system. Their total number is

$$\sum_v g_v + |E| - |V| + 1.$$

The orientability hypothesis yields an orientable handlebody of this genus. Every g_v is nonnegative, and $|E| - |V| + 1$ is the nonnegative cycle rank of a connected graph. Their sum is zero exactly in the stated ball case. □

Corollary 4.14 (A terminal classifier for a marked normal cut)**.** *Take the existing authenticated normal-cut parallelity decomposition. Assume that every core component has a successful certificate from Corollary 4.5, every interval-bundle base has nonempty boundary, and every maximal core-labelled boundary run is proper. Then each component of the entire cut is a handlebody. Its genus is given by (G8), using the core genera from (G4), the bundle genera from (G6), and the actual maximal-run incidence graph. In particular, the entire cut is one three-ball exactly when that graph is connected and acyclic and all its vertex genera are zero.*

Proof. The core certificates give the handlebody pieces. Lemma 4.11 gives the same conclusion for the bundle pieces, including those with nonorientable bases. Lemma 4.12 supplies their separated disk interfaces. Apply Theorem 4.13. □

This classification uses only the existence of the homeomorphisms supplied by the core cancellations. Such a homeomorphism automatically transports the originally authenticated

boundary disks to disjoint embedded disks. Because Theorem 4.13 allows arbitrary disk placements, one need not explicitly transport their parametrizations to compute the genus. Explicit marking transport would again be necessary for a later operation that needs specified curves or disks, rather than this terminal topological type.

An implementation must also preserve the distinction between an actual piece and a multiplicity-weighted profile of pieces. A bundle component with a core port has a source owner and multiplicity one. Unattached repeated profiles can contribute many isolated graph components. The one-ball conclusion requires the total connected-component count to be one, including those unattached components.

4.6 A concrete extension: absorbing disc-base annuli

A full core-labelled boundary circle prevents disk gluing. For a disk base, however, its bundle is a two-handle. Lackenby's refined algorithm already uses this replacement for interior parallelity bundles[3, Section 10, step (2′)]. The contribution below supplies the source-chart incidence row and proves compatibility with the present geometric cancellation certificates.

Proposition 4.15 (Source-aware absorption of a disc-base bundle)**.** *In the authenticated source-parallelity decomposition, let $P \to F$ be a connected bundle component whose base is a disk and whose entire base boundary is core-labelled. Then P can be absorbed into its owner core component as one two-handle. No zero-, one-, or three-handles need be added.*

Use the full ordered boundary port circle and its native source charts. Every port of core dimension one is a face-sector strip running from one end of that source one-handle to the other, and every port of core dimension zero lies in a zero-handle boundary. Thus the new attaching circle has raw unsigned incidence

$$n_h^{\mathrm{new}} = \#\{\textit{ports on this circle of core dimension one and owner } h\}. \tag{G9}$$

After deleting spanning-forest columns, this row may be appended to the existing authenticated core incidence matrix. The geometric pivot, saturation, and source row-leaf theorems then apply to the augmented core. The assertion holds simultaneously for any collection of such bundle components. There are $O(T)$ possible absorbed components and $O(T)$ source port occurrences in all their new rows together.

Proof. An interval bundle over a disk is $D^2 \times I$. Its intersection with the core is, by the full source product-intersection conclusion,

$$p^{-1}(\partial F) = \partial D^2 \times I.$$

This is exactly the attaching region of a three-dimensional two-handle. The annulus is connected, so its image lies in one core component. The two horizontal disks are the free boundary of the attached handle.

To authenticate its attaching circle relative to the core's zero/one structure, choose the midpoint section of its fibers around ∂F. The native fiber transition maps preserve this midpoint, including maps that reverse the fiber. The resulting curve is therefore an embedded circle in the attaching annulus. On a core-zero port it lies in the boundary of the indicated zero-handle. The native constructor's core-one ports come precisely from the `face_sector` sides of selected edge-cap polygons. Each such side traverses the lateral boundary of its core one-handle between the two source tetrahedron ends, even when those ends belong to the same tetrahedron or zero-handle. The midpoint curve consequently meets that one-handle's co-core exactly once. The source face flags, rather than equality of the tetrahedron names, identify these ends. This proves (G9).

There is also an explicit signed chart check, although the saturated classifier needs only unsigned occurrences. Let the oriented source ends of a core one-handle h have flags (t_h, f_h) and (u_h, g_h). For the face-sector port p, inspect the initial and terminal cap-edge endpoint flags in its local direction and set

$$\delta_p = \begin{cases} +1, & \text{initial} = (t_h, f_h), \quad \text{terminal} = (u_h, g_h), \\ -1, & \text{initial} = (u_h, g_h), \quad \text{terminal} = (t_h, f_h). \end{cases} \tag{G10}$$

Require one of these two cases. If the port is traversed with sign s_p in the canonical boundary word, its raw letter is $x_h^{s_p\delta_p}$. A core-zero port stays in its owner zero-handle, while a core-one port moves between the two stated ends. The resulting sequence must be a closed path in the source zero/one graph. Both tetrahedron and face flags are needed: equality of tetrahedron labels alone does not distinguish the two ends of a self-pairing. No abelian signed sum replaces the unsigned count in (G9).

These port charts lie in the boundary of the core zero/one-handle union outside the attaching regions of the original core two-handles. This is part of the genuine source dual-handle structure: the complete lower attachments of complementary handles stay in the core, while the selected product contacts occupy their recorded, disjoint boundary regions. Compatible small collars make the new midpoint circles disjoint from the original two-handle attaching curves. Hence all the new handles can be placed at the same two-handle stage, with exactly the authenticated rows just computed. Tree contraction deletes only the tree entries, as before. The preceding geometric cancellation theorems apply to this augmented handle structure without introducing a three-handle.

Different bundle components are disjoint in the embedded source bundle, so their attaching annuli and midpoint curves can be regularized disjointly. Absorbing them leaves the marked interfaces of the remaining components intact. Each absorbed component contains at least one core port, and every port has one actual owner. The native $O(T)$ port bound therefore bounds both the number of new two-handles and the total number of port occurrences used in their rows. □

This proposition is a proved extension of the conservative classifier; absorption is not performed by the delivered disk-interface consumer. It suggests a direct next implementation: identify the certified base signature $(\chi, b) = (1, 1)$ with its complete boundary core-labelled, append its source-authenticated incidence row, run the core cancellation checker again, and apply disk gluing to the remaining bundle pieces. The base's surface structure and the full port charts must be verified; the two numerical values (χ, b) alone would not supply the necessary geometric conclusion. A bundle over an annulus or a Möbius band is a solid torus rather than a two-handle, so the proposition does not silently extend to those cases.

4.7 Why one ball after a connected cut certifies the unknot

Proposition 4.16 (The one-ball cut criterion)**.** *Let M be a connected compact orientable three-manifold with boundary one torus. Let $S \subset M$ be a connected orientable properly embedded surface with nonempty boundary. If the manifold obtained by cutting M along S is one three-ball, then S is a disk and M is a solid torus. If M is an authenticated knot exterior in S^3, this certifies that the knot is the unknot.*

Proof. The ambient and surface orientability imply that S is two-sided. Let M_S denote the manifold cut along a product neighborhood of S. Euler characteristic under this cut gives

$$\chi(M_S) = \chi(M) + \chi(S).$$

For a compact orientable three-manifold, $\chi(M) = \frac{1}{2}\chi(\partial M)$; this also follows by doubling and using the vanishing Euler characteristic of a closed odd-dimensional manifold. The torus boundary gives $\chi(M) = 0$, while the ball hypothesis gives $\chi(M_S) = 1$. Hence $\chi(S) = 1$. Classification of connected orientable surfaces with nonempty boundary implies that S is a disk.

Reconstructing M from the cut ball inserts a neighborhood of this disk, namely one three-dimensional one-handle attached along its two boundary copies. Orientability makes the result a solid torus.

For a knot exterior, an essential meridian disk of this exterior has boundary equal, up to orientation, to the knot's preferred longitude: that primitive slope generates the kernel of the map from the boundary torus's first homology to the exterior's first homology. Inside the knot's tubular neighborhood this longitude cobounds an annulus with the knot. The disk and annulus form an embedded spanning disk for the knot, proving that it is the unknot. □

The hypotheses must be used literally. Finding a ball among several cut components is insufficient. A cut component of positive handlebody genus is insufficient. A multiple normal vector can describe a disconnected surface, so its connectedness and orientability cannot be inferred from a successful core certificate.

4.8 An all-size Fibonacci family with ball cores

We now prove a uniform success theorem for the native complementary core of the layered solid-torus fixture from report 67 [6]. This gives a construction and proof for every family member; finite timings play no role in the theorem.

Let $F_1 = F_2 = 1$ and $F_{j+2} = F_{j+1} + F_j$. For $t \geq 1$, number the tetrahedra $0, \ldots, t-1$. For each $j < t-1$, glue face 0 of tetrahedron j to the next tetrahedron using the vertex permutation

$$P_0 = (2, 1, 3, 0),$$

and face 1 using

$$P_1 = (0, 3, 1, 2).$$

At the last tetrahedron, pair face 0 to face 1 using $P = (1, 2, 3, 0)$. Thus the unpaired faces are the top faces 2 and 3. For an arbitrary positive integer m, use normal-coordinate rows

$$m\big(F_{t-j+1}, F_{t-j+1}, 0, 0, 0, 0, F_{t-j}\big), \qquad 0 \leq j < t. \tag{G11}$$

The primitive fixture is the standard meridian of this layered solid torus. The following theorem concerns its native *complementary core*, for every positive scale, not the number of components of the whole cut.

Theorem 4.17 (Uniform ball-core certificates for the Fibonacci fixtures)**.** *For every $t \geq 1$ and every positive integer m, the native core of the cut specified by* (G11) *has handle counts*

$$h_0 = 4t, \qquad h_1 = 6t - 3, \qquad h_2 = 2t - 2, \qquad h_3 = 0, \tag{G12}$$

and is a three-ball. With the native source ordering described below, the increasing-identifier spanning tree followed by immutable-source row-leaf deletion removes every two-handle. The certificate has $O(t \log(t+1))$ bits and $O(t)$ source incidences, independently of m and of the numerical normal weights.

Proof. We describe the zero/one-handle graph, enumerate all interior edge links, and exhibit a row-leaf order valid for every t.

The zero/one-handle graph. At tetrahedron j the occupied triangle types are 0 and 1, and the occupied quadrilateral type is column 6. The four exceptional chambers are denoted A_j, B_j, U_j, V_j: the two quad-end chambers, followed by the triangle tips at vertices 0 and 1. In this order their identifiers are $4j, 4j+1, 4j+2, 4j+3$. Applying the two layer permutations to the central face region and its occupied corner regions gives the following one-handles in each nonfinal layer:

identifier	ends
$6j$	(B_j, A_{j+1})
$6j+1$	(V_j, V_{j+1})
$6j+2$	(A_j, U_{j+1})
$6j+3$	(A_j, B_{j+1})
$6j+4$	(U_j, U_{j+1})
$6j+5$	(B_j, V_{j+1}).

(G13)

The last self-pairing gives three more one-handles, of identifiers $6t-6, 6t-5, 6t-4$, with ends

$$(B_{t-1}, A_{t-1}), \qquad (V_{t-1}, B_{t-1}), \qquad (A_{t-1}, U_{t-1}),$$

respectively. These lists already prove $h_0 = 4t$ and $h_1 = 6t-3$.

For $t \geq 3$, the increasing-identifier spanning tree consists of

$$\{0, \ldots, 10\} \cup \bigcup_{j=2}^{t-2} \{6j, 6j+1, 6j+2, 6j+3\}. \tag{G14}$$

Indeed, the first six edges in (G13) form two trees on four vertices each. Edges $6, 7, 8, 9$ attach the four vertices of tetrahedron 2, and edge 10 joins the two previous trees. Thus the first eleven edges form a tree on the first twelve vertices. Thereafter the first four edges of a layer each attach one new vertex to the existing tree. The other two edges of that layer join vertices already connected, as do the three final self-pairing edges. Thus (G14) has exactly $4t-1$ edges and is the claimed spanning tree. When $t = 2$ the tree is $\{0, \ldots, 6\}$, and when $t = 1$ all three one-handles form the tree.

The bottom self-pairing identifies vertices $1, 2, 3, 0$ successively. The layer permutations propagate this identification to every preceding tetrahedron. All source vertices consequently lie in the single boundary vertex class containing the top vertices. There are no interior vertex links and hence no core three-handles.

The complete edge inventory. Write $(j; ab)$ for local edge $\{a, b\}$ in tetrahedron j. For every $0 \leq j \leq t-2$ there is an interior edge E_j represented by $(j; 23)$. For $0 \leq j \leq t-4$, its cyclic list of local occurrences is

$$(j; 23),\ (j+1; 12),\ (j+2; 13),\ (j+3; 01),\ (j+2; 02),\ (j+1; 03). \tag{G15}$$

The exit faces are, in order, $1, 0, 0, 3, 3, 2$. Each transition follows by applying the indicated layer permutation or its inverse. For the penultimate edge, $j = t-3$, the bottom self-pairing shortens the list to

$$(j; 23),\ (j+1; 12),\ (j+2; 13),\ (j+2; 02),\ (j+1; 03), \tag{G16}$$

and for the terminal edge, $j = t-2$, the list is

$$(j; 23),\ (j+1; 12),\ (j+1; 23),\ (j+1; 03). \tag{G17}$$

These cycles are disjoint as sets of local edge occurrences. For $t \geq 3$, the remaining occurrences form exactly the following three boundary classes:

$$\begin{aligned} &\{(0;01)\}, \\ &\{(0;02),(0;13),(1;01)\}, \\ &\{(0;03),(0;12),(1;02),(1;13),(2;01)\}. \end{aligned} \tag{G18}$$

Their transitions again follow directly from the face maps; each class meets a top boundary face. The interior lists use $6t-9$ local edge occurrences and the boundary lists use the remaining nine, so this is the complete edge inventory. For $t=2$, omit $(2;01)$ from the last boundary class; the sole interior cycle is (G17). For $t=1$, the three boundary classes in the only tetrahedron are $\{01\}$, $\{02,13\}$ and $\{03,12,23\}$, and there is no interior edge.

Every E_j has positive normal edge weight, equal to mF_{t-j}. The native core therefore has exactly two two-handles around it, at rank zero and at the full edge weight. Thus there are exactly $2t-2$ core two-handles, proving the remaining count in (G12).

The literal attaching rows. Give these two handles identifiers $2j$ and $2j+1$. Reading their endpoint ranks around (G15) and using (G13) gives the following one-handle crossing lists for $0 \leq j \leq t-4$:

$$\begin{aligned} 2j: &\quad (6j+5,\ 6j+7,\ 6j+13,\ 6j+17,\ 6j+9,\ 6j), \\ 2j+1: &\quad (6j+3,\ 6j+6,\ 6j+14,\ 6j+16,\ 6j+10,\ 6j+2). \end{aligned} \tag{G19}$$

For $j=t-3$, the shortened link gives

$$\begin{aligned} 2j: &\quad (6j+5,\ 6j+7,\ 6j+13,\ 6j+9,\ 6j), \\ 2j+1: &\quad (6j+3,\ 6j+6,\ 6j+14,\ 6j+10,\ 6j+2), \end{aligned} \tag{G20}$$

and for $j=t-2$ it gives

$$\begin{aligned} 2j: &\quad (6j+5,\ 6j+7,\ 6j+6,\ 6j), \\ 2j+1: &\quad (6j+3,\ 6j+6,\ 6j+8,\ 6j+2). \end{aligned} \tag{G21}$$

For example, the two endpoint ranks at a face arm select its corner-end and central exceptional regions respectively; applying the face permutation selects the corresponding endpoint in (G13). The cyclic edge lists enumerate every such passage, so there are no additional crossings. All entries in the displayed lists have multiplicity one. The numerical values of the full edge ranks do not change which exceptional endpoint is selected.

A uniform source row-leaf order. Assume $t \geq 3$. Name the non-tree columns

$$\begin{aligned} &a_j = 6j+5, && 1 \leq j \leq t-2, \\ &b_j = 6j+4, && 2 \leq j \leq t-2, \\ &x = 6t-6, \quad y = 6t-5, \quad z = 6t-4. \end{aligned} \tag{G22}$$

Use a_0 and b_1 as formal names for absent columns: their identifiers 5 and 10 belong to the spanning tree. Deleting the tree entries from (G19)–(G21) gives the following support table, where absent formal columns are omitted:

range	row $2j$	row $2j+1$
$0 \leq j \leq t-4$	$\{a_j, a_{j+2}\}$	$\{b_{j+1}, b_{j+2}\}$
$j = t-3$	$\{a_{t-3}, y\}$	$\{b_{t-2}, z\}$
$j = t-2$	$\{a_{t-2}, x, y\}$	$\{x, z\}$.

(G23)

This is a square matrix of size $2t - 2$.

First peel the odd rows in increasing order. Since b_1 is absent, the successive deleted columns are

$$b_2, b_3, \ldots, b_{t-2}, z, x.$$

The initial range is empty when $t = 3$; the same rule still deletes z and then x.

After x is removed, view the even rows as edges on the formal vertices $a_0, a_1, \ldots, a_{t-2}, y$. The first range of rows supplies the parity-step edges $a_j a_{j+2}$. The last two even rows join the final vertices of these two parity chains to y. Their union is therefore a single path, with one endpoint the absent column a_0. Peel along this path from that endpoint. The deleted columns are the even-indexed a's in increasing order, followed by y, followed by the odd-indexed a's in decreasing order, ending at a_1. Every row is a source row leaf when it is used. Thus all even rows are removed as well.

For $t = 2$, the two surviving columns have identifiers 7 and 8, and the two reduced rows form the identity matrix. For $t = 1$, the reduced matrix is empty. Theorem 4.8 now proves that the core is a ball for every t.

Finally, multiplying (G11) by a positive integer changes neither its occupied disk types nor the positivity of any relevant edge weight. Every exceptional chamber, endpoint rank, source crossing list and leaf order above therefore remains unchanged for all $m \geq 1$. There are $O(t)$ forest edges and pivots, and each crossing list has bounded length, proving the certificate-size assertion. □

The primitive coordinates in this family already have exponential numerical weight in t, whereas the core certificate is linear in its number of source records. The theorem establishes an exact, nontrivial independence from normal multiplicity for this core decision. It does not identify a coordinate's numerical magnitude with its binary input length, and it does not extend the one-ball cut criterion to disconnected multiples. The marked interval-bundle assembly remains necessary to classify the entire cut.

5 A complete certificate for the accepted cut class

We now combine the previous constructions into a single conservative consumer. This is a genuine topology operation on a supplied source cut: both the exceptional core and all bundle components contribute to the answer.

5.1 Algorithm and proof

For each supplied vector, the delivered algorithm performs the following steps.

1. Run the established source parallelity construction and its four orbit queries. Retain the geometric core, base profiles, and source ports.
2. Reject the attempt as unresolved if the core has a three-handle. Contract an increasing-ID spanning forest in its zero/one graph. Try the source leaf order on the raw two/one incidence matrix. If this stalls and the matrix is square, the full raw ball-cancellation class is exhausted. Otherwise try a deterministic saturated-count pivot schedule. A stalled schedule returns `INCONCLUSIVE`.
3. Recover the ordered boundary skeleton using the two new queries. Reject the attempt as unresolved if any core attachment is a full boundary circle or any bundle base is closed.
4. Construct one vertex for each certified core component, and one for each ported base component. Give them their certified genera and join them by one edge for each *maximal*

attachment interval. Compute the genus of each connected graph component by disk gluing. Add unported bundle profiles using their binary multiplicities.

5. Check equation (1), then emit the histogram $\{(g, m_g)\}$ and the complete source, orbit, cancellation, and graph proofs.

The order of cheap rejection and interface discovery is a producer choice. The independent checker validates the complete proof regardless of that choice. Failed attempts are not certificates that the cut is not a handlebody union. In particular, the current wrapper deliberately leaves the annular absorption theorem in the geometry section as a further implementation opportunity, rather than silently adding it to measured coverage.

Theorem 5.1 (Sound exact binary handlebody census)**.** *On an input satisfying the source contract, every accepted* `normal-cut-handlebodies-v1` *certificate proves that $M \setminus\setminus S$ is a disjoint union of handlebodies. Its returned multiplicity m_g is exactly the number of components of genus g. No operation needs to enumerate the m_g separate components or their normal disks.*

Proof. The source and boundary checkers establish the actual decomposition $K \cup \mathcal{B}$, the component profiles, and the maximal components of A. The geometric cancellation checker turns each component of K into its certified handlebody. Each bordered bundle is an orientable handlebody of genus $1 - \chi$ of its base. Every retained attaching component is a proper interval in a base boundary and hence lifts to a genuine embedded disk. The source regularization statement makes distinct attachment disks disjoint, including their relevant corner collars.

Consequently the connected components containing the core are exactly those of the finite attachment multigraph. The disk-gluing theorem gives

$$g(C) = \sum_{v \in V(C)} g(v) + |E(C)| - |V(C)| + 1.$$

It applies also to repeated edges: a second disk between already connected pieces contributes a cycle and increases the genus by one. A ported profile has multiplicity one, so no vertex represents omitted copies. Conversely, an unported bundle meets no core component and is an isolated component of the cut; its source multiplicity must be added unchanged. This accounts for every source piece and proves the claimed histogram. The final Euler identity is a redundant arithmetic check, not the topological justification. □

Corollary 5.2 (Exact all-ball predicate on successful certificates)**.** *For an accepted certificate, the entire cut consists of balls if and only if every nonzero histogram entry has genus zero. The entire cut is one ball if and only if the histogram is exactly $\{(0, 1)\}$.*

The second condition, combined with an independently verified connected orientable surface having nonempty boundary and a one-torus source boundary, has the terminal implication proved in the geometry section. A knot verdict additionally requires the existing diagram-to-exterior provenance chain. Neither surface connectedness nor diagram provenance is inferred from the binary number of balls.

5.2 Time, encoded output, and what has actually been removed

Let E bound the total event count in the six orbit traces and let L be the encoded source size. The interface section gives polynomial bounds for weight runs, tagged selectors, and their integer bit lengths. With the maintained polynomial AHT scheduling theorem, all source/profile/boundary work is polynomial in T and B. The new boundary phase explicitly

uses the classical AHT periodic rule; inherited profile queries retain their already justified native scheduler.

There are $O(T)$ core incidences and at most $O(T)$ pivots. Saturated replay uses at most $O(T^3)$ elementary operations on the three-element semiring. For a source leaf order, the elimination portion visits each incidence only a constant number of times, with heap overhead in the producer. The current forest verification and component bookkeeping can take $O(T^2)$; the implementation should not be advertised as a fully linear core checker. The finite attachment graph is also of size $O(T)$, with polynomial-time bookkeeping. Binary histogram arithmetic is charged in L.

Corollary 5.3 (Local bit-polynomial terminal operator)**.** *Without an artificially exhausted work budget, the delivered terminal operator finishes in bit-polynomial time in the supplied triangulation and normal-vector encoding. Its successful certificates have polynomial encoded size and polynomial replay cost in that encoding.*

A supplied successful raw schedule and disk-interface decomposition are checkable; this does not claim that every handlebody triangulation satisfies the criterion. For square cores the source leaf theorem does give a complete producer within the raw cancellation class for the fixed source forest. The delivered positive-genus fallback has no such completeness claim. The stronger exhaustive survivor-set result in Proposition 4.10 supplies a theoretical $\binom{k}{g}\,\mathrm{poly}(T)$ alternative, and is quasi-polynomial for $g = O(\log T)$, but that exhaustive search is not implemented here.

The distinction between an endpoint bound and a certificate-size bound is visible in the large tests. The number of boundary words can remain fixed while the proof grows substantially with coordinate bit length. At eight tetrahedra and multiplicity 2^{16384} the boundary-only certificate is about 31 MB. The entire source/census certificate is larger still. No claim of constant memory or constant-size proofs follows from having three interface intervals.

For a surface with m parallel meridian disks, an explicit list of the m cut components already requires $\Omega(m)$ output positions. If $m = 2^b$, this is exponential in the extra $b+1$ bits of multiplicity. Returning "m balls" avoids that output barrier. On successfully certified inputs this is an asymptotic change of representation and operation, even when a small explicit C++ computation is faster in wall-clock time.

6 A complementary acceleration: coordinate blocks in normal-sector search

The marked-interface results address what must be preserved after cutting. This section concerns an earlier cost: finding normal-coordinate candidates inside one supplied compatible quadrilateral sector. It does not assume that the sector is complete for disc discovery. The contribution is an exact decomposition of the matching search, a source-certified implementation, and a genuine solid-torus family on which the maintained exhaustive intersection iterator performs exponentially more candidate solves than the decomposed iterator. The linear-algebra principles are classical; the source-level decomposition, its interaction with canonical triangle minima, and the explicit geometric separation are the contributions developed here.

The comparison is with the repository snapshot identified earlier [6]. Its current modules already contain support contraction, matching-sign implications, grouped Euler minima, generic Q-corner screening, and special algorithms at matching nullity at most three. Reports 67 and 75 also retain anchor-LP and exact witness-reuse approaches. None of those results is being claimed again as a new theorem. The present interface is additive and opt-in; it does not change the recognizer's default dispatch.

6.1 Matching-coordinate factors and canonical Euler characteristic

Fix a source accepted by the native finite-manifold validator: a connected, compact, orientable triangulated three-manifold with one torus boundary and sphere or disc vertex links. Let T be its tetrahedron count. A compatible sector allows at most one quadrilateral type per tetrahedron. After eliminating triangle matching, write

$$L = \ker A \subseteq \mathbb{R}^k, \qquad C = L \cap \mathbb{R}^k_{\geq 0},$$

where the k coordinates are the allowed quadrilateral types. A coordinate block is a subset I of these indices; $\mathbb{R}^I$ means vectors supported on I. All spaces and eliminations below are over the rationals when implemented.

Theorem 6.1 (The finest coordinate decomposition)**.** *There is a polynomial-time exact construction of pairwise disjoint blocks $I_1, \dots, I_s$ and a set Z of coordinates identically zero on L such that*

$$L = \bigoplus_{j=1}^{s} L_j, \qquad L_j = L \cap \mathbb{R}^{I_j}, \qquad C = \bigoplus_{j=1}^{s} C_j, \quad C_j = L_j \cap \mathbb{R}^{I_j}_{\geq 0}.$$

The nonzero blocks refine every other coordinate direct-sum decomposition. Every extreme ray of C belongs to exactly one nonzero C_j, and every extreme ray of C_j is an extreme ray of C.

Proof. Compute a canonical kernel basis $r_1, \dots, r_d$ with free coordinates $f_1, \dots, f_d$ satisfying $(r_i)_{f_j} = \delta_{ij}$. The native gauge uses the rightmost independent free coordinates; any fixed canonical gauge works. Join coordinates that occur together in the support of a basis row. The connected components of this graph on occurring coordinates are the I_j; the nonoccurring coordinates form Z. Each basis row is supported in one component. Grouping the basis rows therefore proves the direct-sum formula for L, and coordinatewise nonnegativity proves the formula for C.

For the refinement assertion, consider any coordinate decomposition of L. Project r_i onto one of its summands. Except in the summand containing f_i, this projection has every free coordinate zero. It must therefore vanish, because restriction to the free coordinates is injective on L. Thus every support of r_i lies in a single summand, and so does every connected component constructed above. This argument also explains why one must first canonicalize an arbitrarily mixed input basis.

A nonzero vector with nonzero components in two of the C_j decomposes as a sum of nonproportional feasible vectors and cannot generate an extreme ray. Conversely, if a vector supported in I_j is decomposed into two nonnegative vectors in C, both summands vanish outside I_j. Its local extremality therefore implies global extremality. Rational elimination and the support graph have polynomial bit complexity in their input encoding. □

This is an algebraic cone decomposition. It does not claim that the physical components of a normal surface are disjointly partitioned by these blocks. The earlier source-equation support decomposition in report 69 has a different, geometric purpose. Cancellation in eliminated matching rows cannot by itself establish physical disjointness.

Let $X(q)$ be the canonical standard-coordinate lift: subtract the least triangle potential at each global vertex. Write $p_c(q)$ for its linear corner potentials, G_v for the corners at vertex v, and $\lambda_v = \chi(\operatorname{link} v)$. The exact source Euler compiler gives

$$F(q) := \chi(X(q)) = \gamma \cdot q - \sum_v \lambda_v \min_{c \in G_v} p_c(q). \tag{2}$$

Here λ_v is one for a boundary vertex and two for an interior vertex.

Lemma 6.2 (The canonical-lift defect)**.** *For $q, r \in C$, put*

$$\delta_v(q,r) = \min_c(p_c(q) + p_c(r)) - \min_c p_c(q) - \min_c p_c(r).$$

Then $\delta_v(q,r) \geq 0$ and

$$X(q) + X(r) = X(q+r) + \sum_v \delta_v(q,r) L_v,$$

where L_v is the standard-coordinate vertex-link vector. Consequently, if every $\lambda_v \geq 0$, then

$$F(q+r) \leq F(q) + F(r), \qquad F(aq) = aF(q) \quad (a \geq 0).$$

Proof. Every sum $p_c(q) + p_c(r)$ is at least the sum of the two separate minima. Subtracting the three canonical triangle formulas gives the displayed coordinate identity; quadrilateral coordinates agree identically. Applying the source Euler covector gives $F(q+r) = F(q) + F(r) - \sum_v \lambda_v \delta_v(q,r)$. The asserted inequality and positive homogeneity follow. □

Theorem 6.3 (Exact block localization of the Euler maximum)**.** *Assume $C \neq \{0\}$ and every $\lambda_v \geq 0$. Omit blocks whose feasible cone is zero. Then*

$$\max_{q \in C,\, \|q\|_1 = 1} F(q) = \max_j \max_{q \in C_j,\, \|q\|_1 = 1} F(q).$$

In particular, the sector contains a canonical positive-Euler normal surface if and only if one coordinate block does. This remains true when triangle minima couple several matching blocks.

Proof. The normalized sections are compact. For a normalized q, decompose it as $q = \sum_j q_j$ using Theorem 6.1. Put $w_j = \|q_j\|_1$ and, when $w_j > 0$, $u_j = q_j / w_j$. The preceding lemma gives

$$F(q) \leq \sum_j F(q_j) = \sum_{w_j > 0} w_j F(u_j) \leq \max_j \max F|_{C_j \cap \{\|q\|_1 = 1\}},$$

since $\sum_j w_j = 1$. Every local normalized section is contained in the global section, proving the reverse inequality. Rational polyhedral vertices suffice for the maxima of the convex piecewise-linear function; clearing denominators produces integral normal coordinates. □

The sign condition is substantive. For the abstract model $C = \mathbb{R}^2_{\geq 0}$ and $F(x,y) = \min(x,y)$, corresponding to a negative minimum weight, both coordinate axes have value zero but the interior has positive value. This model is excluded by the source hypotheses.

Positive Euler is still not a disc predicate. A positive ray may contain a sphere or a boundary-inessential disc, and it need not be the ray required by a downstream topological test. Thus the theorem permits a complete Euler exclusion or an Euler witness; it does not permit declaring an essential-disc search exhausted after testing only the block Q rays.

6.2 When complete standard rays also split

For each global vertex v, choose an anchor $a_v \in G_v$. Say that v varies on a matching block I_j if some restricted difference $(p_c - p_{a_v})|_{L_j}$ is nonzero. Connect any two blocks on which the same vertex varies, and take connected components of this graph. Call their unions $J_1, \ldots, J_u$. These definitions depend only on differences of potentials, so they are invariant under additive linear changes of the potential gauge.

Theorem 6.4 (Standard rays and minimum-interaction clusters)**.** *The non-link standard extreme rays of the full sector are exactly the union of the non-link standard extreme rays of its sectors restricted to $J_1, \ldots, J_u$. Every such ray occurs in one restricted sector only.*

Proof. Write $q = \sum_i q_i$ according to these unions. At a vertex v, all nonconstant differences $p_c - p_{a_v}$ vary on at most one J_i. After the common anchor term cancels in canonical normalization, the triangle coordinates at v therefore depend on q_i alone; on every other summand they are zero. If no block varies at v, all canonical triangle coordinates there are zero. It follows coordinatewise that

$$X(q) = \sum_i X(q_i).$$

Every non-link standard extreme ray is canonical: a positive vertex-link offset would split it into a nonzero canonical vector and a link vector. The displayed identity then forces its quadrilaterals into one J_i. Conversely, forbidding every Q coordinate outside J_i defines a coordinate face of the full standard cone. An extreme ray of that face is extreme in the full cone, and its canonical lift is the same source lift. Different unions have disjoint Q supports, preventing duplicates. □

Let $k_j = |I_j|$, $d_j = \dim L_j$, and $h_j \le k_j$ be the number of distinct projected Q coordinate hyperplanes. Complete Q enumeration now attempts

$$\sum_j \binom{h_j}{d_j - 1} \tag{3}$$

candidate intersections, with zero-dimensional blocks omitted. The same argument handles feasible cones of smaller dimension than their matching spaces. Each candidate has polynomial bit cost; exact rational coefficients are controlled by determinants of input-sized matrices. Source preparation, projection, actual coordinate output, and certificate replay must all be charged. For complete standard enumeration, the corresponding sum uses the coarser J_i, with at most $O(|J_i|^2)$ potential hyperplanes per cluster; the maintained low-dimensional envelope algorithms can be retained.

Consequently logarithmic *maximum local* nullity gives a quasi-polynomial supplied-sector bound even if total nullity is linear. No theorem here bounds that maximum on arbitrary knot exteriors or supplies a complete small family of sectors. A single large cluster can retain the original exponential cost.

6.3 A genuine solid-torus family with exponential iterator separation

The artifact constructs a simplicial solid torus by triangulating $\Delta^2 \times S^1$, with the circle divided into three edges. Each triangular prism has its three staircase tetrahedra, giving nine tetrahedra and embedded boundary triangles. Attach a sequence of triangulated bipyramid balls along embedded boundary triangles. Each attachment preserves the solid torus. The outgoing face in the construction contains two newly introduced vertices and one existing vertex, so it is again an embedded triangle suitable for the next attachment.

For the measured family, one ball has tetrahedra

$$(D, E, A, B), \qquad (D, E, B, C), \qquad (D, E, C, A),$$

with cyclic internal face pairs. Allow repository quadrilateral types $(1, 1, 2)$ in those tetrahedra, and no quadrilaterals in the initial torus.

Proposition 6.5 (Independent bipyramid matching factors)**.** *With b attached three-tetrahedron balls, the source has $T = 9 + 3b$, $k = 3b$, and $d = 2b$. Its Q cone is the direct product of b copies of*

$$\{(u_0, u_1, u_2) \ge 0 : u_0 + u_1 = u_2\}.$$

Each factor has the canonical matching basis $(-1,1,0),(1,0,1)$ *and its two extreme rays are* $(1,0,1)$ *and* $(0,1,1)$*. The global Q cone therefore has exactly* $2b$ *rays.*

Proof. At the vertex D, the three internal face equations for triangle potentials z_0, z_1, z_2 read

$$z_0 + u_0 = z_1, \qquad z_1 + u_1 = z_2 + u_2, \qquad z_2 = z_0.$$

Their sum is $u_0 + u_1 = u_2$. The cycle at E gives its negative; the equatorial vertex graphs are trees and supply no other Q equation. Gluing the ball along a boundary triangle joins three pairs of distinct vertex graphs by one edge each. These are bridges, so they create no additional triangle-potential cycle or Q relation. This proves independence inductively. The claimed basis, dimension, and rays follow directly from the displayed nonnegative equation. □

Corollary 6.6 (Exponential-to-linear candidate separation)**.** *On this family the maintained exhaustive global Q iterator attempts exactly*

$$\binom{3b}{2b-1} = \Theta\left(\frac{(27/4)^b}{\sqrt{b}}\right)$$

candidate intersections. The block iterator attempts exactly $3b$ *and returns the identical* $2b$ *rays. Its complete construction and independent block-coverage replay have polynomial bit complexity in* b*.*

Proof. The three coordinate forms in each two-dimensional factor define three distinct hyperplanes; forms on different factors are not proportional. The global iterator explicitly visits every $(2b-1)$-subset of these $3b$ hyperplanes. The new iterator visits the three one-hyperplane candidates in each factor. The kernel and output coordinates in this family have polynomial encoding length, and the independent verifier makes only local fixed-dimensional ray reconstructions after its polynomial global source and splitting checks. Stirling's formula gives the asymptotic count. □

This is a separation for the maintained iterator, not an exponential lower bound for the geometric decision problem or for every ray-enumeration algorithm. In the triangular family, one local nonnegativity inequality is redundant. The same construction admits a stronger variant with a four-tetrahedron bipyramid ball and types $(1,1,2,2)$: its relation is $u_0 + u_1 = u_2 + u_3$, its four coordinate inequalities are irredundant facets, and it has four Q rays. For b such balls, the same proof gives

$$k = 4b, \quad d = 3b, \quad \text{global candidates} = \binom{4b}{3b-1}, \quad \text{block candidates} = 6b, \quad \text{Q rays} = 4b.$$

Thus removing redundant inequalities does not remove the general independent-factor obstruction to global subset enumeration.

6.4 A source-realized counterexample to splitting standard rays too far

The two-ball triangular source already distinguishes the two theorems. Put

$$q = (x, y, x+y, z, w, z+w), \qquad x, y, z, w \geq 0.$$

Its four Q rays are the four coordinate axes in these parameters. The source's nonconstant vertex potential groups, after anchor shifts and duplicate removal, have forms

$$\{0, y\} \text{ (twice)}, \quad \{0, -z\} \text{ (twice)}, \quad \{0, -x\}, \quad \{0, w, w-x\}.$$

Only the last group changes its minimizing form inside the Q cone; its minimum is $\min(0, w-x)$. The canonical standard fan is therefore the nonnegative orthant subdivided by $x = w$. It has five rays, adding $e_x + e_w$ to the four coordinate axes. The extra standard ray has

$$q = (1, 0, 1, 0, 1, 1).$$

It is extreme because the independent conditions $y = 0$, $z = 0$, and $x = w$ leave one feasible dimension. These exact source potential groups and the five-ray list are also reconstructed by the independent dense verifier. The minimum-interaction construction merges the two matching blocks, as required. A proposed implementation that enumerated standard rays in the separate Q blocks would omit this genuine ray.

6.5 Implementation, verification, and finite evidence

The new producer is `fastunknot/sector_blocks.py`. It exposes complete Q and standard enumeration, complete ray-set certificates, and a canonical positive-Euler screen. Its companion `sector_blocks_verify.py` imports neither the block producer nor its support-graph construction. The checker reconstructs the dense standard matching model from the original source. For each submitted block and each independently computed kernel basis vector, it checks that the coordinate projection still satisfies all Q equations. It checks that every omitted coordinate vanishes on the full kernel. These identities prove the advertised direct-sum coverage without trusting a block count. For standard certificates it additionally checks that each vertex varies on at most one submitted cluster. It then reconstructs every local ray using the existing independent dense enumerator.

The positive-Euler certificate verifies matching, permitted types, canonicality, and positive Euler directly; it asserts no component topology. A negative Euler certificate includes complete local Q exhaustions and the checked splitting. Missing blocks, incorrect fine partitions, missing or repeated rays, altered source identities, and noncanonical positive vectors are rejected. Candidate limits yield an inconclusive result without an exhaustion certificate; callback interruption propagates. Existing signed-hexadecimal integer transport is retained for ray entries.

The finite audit covers all 188 retained source sectors of matching nullity four or five under the same support-generation policy used by the maintained generic Euler tests. All new Q ray sets equal the old complete sets, and all Euler decisions equal the archived complete standard-coordinate oracle. Independent replay accepts every submitted certificate. There are 180 split sectors, but only 19 have fewer candidate intersections; the total drops from 1,275 to 1,110, with maximum individual factor 10/3. Euler outcomes remain 12 exclusions and 176 positive results. These are sector counts, not counts of distinct knot types or new recognition cases.

Nine focused test methods additionally cover dense changes of kernel basis, zero cones, both enumeration phases on all 64 assignments of a three-tetrahedron source, the genuine mixed standard ray, malicious certificate mutations, integer transport, and cancellation. The source family has also been checked through eight triangular bipyramid attachments. These finite checks supplement the proofs rather than replacing their infinite claims.

b	T	Old candidates	Block candidates	Old seconds	Block seconds	Paired gain
1	12	3	3	0.006644	0.006247	1.054
2	15	20	6	0.016219	0.013247	1.217
3	18	126	9	0.051434	0.021766	2.363
4	21	792	12	0.267364	0.032702	8.134
5	24	5005	15	1.848340	0.044701	39.974

The timing task includes complete source preparation, Q-ray production, independent complete ray-set replay, and canonical ray-list hashing. The new arm additionally serializes its block certificate. Each row uses three rounds with alternating arm order; times are arm medians and gains are medians of paired ratios, which need not equal ratios of the displayed medians. Small cases are near parity and should not be read as strong speed claims. Larger cases exhibit the predicted avoided intersection work. The raw measurements and exact ray hashes are retained in `sector_block_research/results/benchmark.json`; the 188-case audit is beside it. These are complete supplied-sector Q-enumeration measurements, not end-to-end knot-recognition timings.

The next mathematical targets are more specific than a bound on total nullity. One can ask whether certified geometric reductions control the largest surviving matching block; whether source-side forced-zero proofs expose smaller blocks than the full linear kernel does; and whether a small separator in the minimum-interaction graph admits an exact bounded interface representation without expanding binary normal coordinates. A separate boundary-aware theorem is required to preserve essential-disc information when the Euler screen finds positive but topologically unhelpful rays. Those obligations remain open here.

7 Implementation, independent validation, and measured performance

The delivered implementation adds a certified terminal operation on a supplied normal cut, together with the separate sector decomposition of Section 6. Its computational evidence addresses three different questions: whether the certificates survive independent replay, which supplied cuts satisfy the current sufficient criterion, and how the cost of the compressed operation compares with constructing and recognizing an explicit cut. The experiments do not measure a complete run from a knot diagram to a knot verdict.

7.1 Additive modules and verification boundaries

The contribution consists of eight new modules under `fast/fastunknot/`, arranged as four producer/checker pairs. Existing recognition defaults and existing production modules are unchanged.

Operation	New modules
Ordered interfaces	`parallelity_boundary_words.py`
	`parallelity_boundary_words_verify.py`
Geometric core cancellation	`core_handlebody.py`
	`core_handlebody_verify.py`
Whole-cut census	`normal_cut_handlebodies.py`
	`normal_cut_handlebodies_verify.py`
Sector factors	`sector_blocks.py`
	`sector_blocks_verify.py`

Table 3: Additive implementations and their separate certificate checkers. The research drivers, fixtures, and tests are additional artifacts.

The interface checker reconstructs its geometry through the existing raw-cut-derived reference implementation. It builds the two orbit systems independently, invokes the established weighted-orbit checker, and reconstructs the signed words and maximal attaching intervals without calling the new producer's query builder or traversal. The core checker obtains unsigned

incidences from authenticated handle cycles; it accepts no supplied incidence summary. It verifies the spanning forest and either the immutable leaf order or the saturated geometric pivot sequence. The whole-cut checker first authenticates both source decompositions and then independently recomputes the attachment graph, its connected components, and its genus histogram.

This independence is an implementation boundary, not a claim that the checkers have no shared foundations. They deliberately reuse established source validation, exact integer transport, and orbit-proof verification. The geometric arguments in Sections 3 and 4 justify the interpretation of the checked data. Tests that disable the producer functions ensure that certificate acceptance does not accidentally call the discovery routines.

The interface schema is `parallelity-boundary-words-v1`; the conditional core schema is `normal-core-handlebody-v1`; and the source-bound whole-cut schema is `normal-cut-handlebodies-v1`. For the last of these, `COMPLETE` asserts the exact binary multiplicity census of the supplied cut. `INCONCLUSIVE` asserts neither non-handlebody topology nor knottedness. A core proof by itself remains conditional on the authenticity of its geometric handle charts.

The production allowance `max_cycles` is shared by the four inherited parallelity queries and the two new interface queries. Exhausting it leaves the attempt unresolved, without a partial accepted census. The checker can cap the total number of supplied orbit operations through `max_operations`; cooperative cancellation also reaches the finite core and graph computations. These are work controls, not wall-clock guarantees. Large integers use the repository's exact hexadecimal JSON transport where needed. The separate sector schemas and their more limited Euler and ray-enumeration meanings are described in Section 6; none silently asserts an essential disc.

7.2 Regression tests and adversarial checks

The recorded integrated run passed **1,644 tests in 346.562 seconds**; the log is `integrated-tests.txt`. Focused validation also passed the 13 core/census methods in `core-census-tests.txt` and the eight boundary methods in `boundary-words-tests.txt`. These focused counts overlap the integrated suite and are not added to it.

The tests check source relabellings, one-sided normal surfaces, empty cuts, parallel meridians, interior-vertex obstructions, repeated incidences, exact large integers, shared budgets, cancellation, and producer-disabled replay. Adversarial mutations alter source fingerprints, endpoint tags, fibre transport, signed words, attachment types, core owners, forest labels, cancellation schedules, and final genera. The expected outcome is rejection of the altered certificate.

Two finite algebraic audits target the central cancellation mechanisms. For 1,200 deterministic random incidence systems, every produced saturated schedule is replayed against unbounded nonnegative integer counts. For 750 deterministic random square matrices of orders one through five, the leaf producer's success is compared with a directly computed permanent. The latter test exposed a stale-queue bug during development: a row already queued as a leaf could cease to be eligible after another column was removed. The producer now rechecks both row survival and its current unique unit entry when popping the queue. This correction preceded the final passing runs. The permanent computation is a test oracle for these small matrices, not an operation used by the classifier.

The boundary tests cover 193 source/vector cases, of which 182 are compared with a literal expanded polygon-boundary oracle. That oracle walks actual boundary edges and reconstructs the corner/fibre identifications separately from the compressed word routine. Eleven large

cases use independent compressed replay and source consistency checks instead of exceeding that oracle's expansion guard. Section 3.8 gives the breakdown and the separate interface-scale measurements. The nine focused sector test methods, their dense independent ray comparisons, the 188-sector audit, and the sector timing experiment are reported in Section 6.

7.3 A source-cut census with an independent topological oracle

The file `census-audit.json` records all 106 supplied cuts in `marked_parallelity_research/corpus.json`, tied to source revision `d2c5e1707fe783bbed06743ac732d70d83e6f27b`. Every native success was independently replayed. For manageable coordinates, a separate Regina path constructs the triangulation from the source face pairings, constructs the normal surface, calls `cutAlong()`, separates the resulting components, and calls `recogniseHandlebody()` on each component. It uses no native core, attachment graph, or genus calculation.

Recorded outcome	Cases
Native `COMPLETE`, with successful independent replay	30
Native `INCONCLUSIVE`	76
Full-circle core attachment	60
Raw cancellation stalled	6
Core has three-handles	10
Explicit Regina oracle completed	99
Native successes compared with Regina; all agreed	23
Binary cases above the explicit expansion cap	7

Table 4: Whole-cut audit. Oracle rows describe a second view of the same 106 cases and are not additional cases. All seven expansion-capped cases have successful native certificates.

The corpus oracle refuses expansion above 4,096 normal disks, before constructing a Regina cut; its subprocess also has a per-case timeout. The recorded run completed all 99 uncapped oracle computations and encountered no oracle timeouts. The seven binary cases were explicitly marked `NOT_ATTEMPTED_EXPANSION_CAP`. They were checked through native certificate replay, without attempting an overflowing or unrepresentable explicit cut and without counting the skip as an oracle agreement.

The 30 successful censuses and 23 independent topological agreements are finite implementation evidence for the proved criterion. The 76 unresolved cases show that the criterion remains restrictive on this selected corpus. In particular, a circle obstruction is a failure of the delivered disk-interface consumer, not a claim that the actual cut is not a union of handlebodies.

The separate `annular-absorption-diagnostic.json` examines precisely those 60 circle-obstructed cases in light of Proposition 4.15. In 38 cases every obstructing full-circle attachment belongs to a disk base; the other 22 have at least one circle on a non-disc base. Fifteen additional boundary queries needed by the diagnostic were independently replayed. This is an eligibility diagnostic: it did not append the proposed new two-handles, run augmented cancellation, or accept an additional census. Consequently the measured coverage remains 30 of 106, rather than 68 of 106.

7.4 Paired timings of the complete supplied-cut operation

The benchmark environment recorded in `census-benchmark.json` is Python 3.12.14, Regina 7.4, and Linux on x86-64. Timings use `time.perf_counter`. Both arms start from the same face pairings and the same supplied standard normal vector in a Fibonacci layered solid torus.

The vector is the primitive meridian vector multiplied by m. Every timed case is checked to give exactly m ball components.

The native arm includes certificate production and independent source-bound replay. The Regina arm includes construction of its triangulation and normal surface, explicit cutting, component separation, and handlebody recognition. JSON serialization and hashing are outside the native timer. Each case has an initial correctness/warm-up execution of both arms, followed by five paired timing rounds. The order within each pair is shuffled using the recorded seed 20261012.

For round r, write n_r for native production plus replay and e_r for the explicit Regina operation. The two time columns below are their separate medians. The comparison statistic is

$$R = \mathrm{median}_{r=1,\dots,5} \frac{e_r}{n_r}, \qquad R \neq \frac{\mathrm{median}_r\, e_r}{\mathrm{median}_r\, n_r} \quad \text{in general.}$$

Thus $R > 1$ favors the native arm. Each case also has three additional Regina-versus-Regina pairs; the final column gives the median of their first-run/second-run ratios. These A/A controls describe timing variability, not a second algorithmic comparison.

T	m	D	T_{cut}	Native median (s)	Regina median (s)	R	A/A
1	1	3	31	0.001794	0.000341	0.205	1.128
1	16	48	226	0.006797	0.001470	0.217	1.051
1	256	768	3,346	0.007451	0.022121	2.922	0.965
1	4,096	12,288	53,266	0.007727	0.429135	54.974	0.989
4	1	29	208	0.183885	0.005508	0.030	1.064
4	8	232	1,013	0.486196	0.026918	0.054	0.980
4	64	1,856	7,453	0.458985	0.182686	0.394	0.968
8	1	228	1,093	1.665820	0.171979	0.105	1.009
8	8	1,824	7,393	2.325465	1.145137	0.499	0.994
12	1	1,592	6,573	4.704264	8.618704	1.817	1.058

Table 5: All ten paired whole-cut benchmark cases. D counts normal disks; T_{cut} is the number of tetrahedra in Regina's explicit cut. Each arm median and the median paired ratio are computed from the saved individual samples.

The results show both overhead and useful savings. Regina is faster in seven of these ten cases, including the small one-copy inputs at $T = 1, 4, 8$. The native operation becomes favorable on the one-tetrahedron repeated-disc examples at $m = 256$ and $m = 4096$, and on the $T = 12$, $m = 1$ input. The largest recorded paired ratio is about 54.97 at $(T, m) = (1, 4096)$; its explicit cut contains 53,266 tetrahedra. At $(12, 1)$, the corresponding ratio is about 1.82, with both native proof phases included.

These finite comparisons support the practical value of avoiding explicit normal-stack expansion on suitable inputs. They do not assert a universal speed advantage. Five samples per arm are descriptive measurements; no CPU affinity, isolated memory environment, or peak-memory measurement was established for this run. The saved A/A data also include noticeable short-duration variation. No whole-recognizer runtime or empirical quasi-polynomial exponent is inferred from this table.

The benchmark has a separate explicit limit of 50,000 normal disks. It is deliberately distinct from the audit's 4,096-disk guard: the largest timed explicit example has $D = 12,288$ and is allowed here. Neither limit is a bound on the number of tetrahedra in the constructed cut.

7.5 Binary capacity and actual certificate sizes

Four additional measurements run only the native producer and independent checker, with multiplicity $m = 2^p$. All return and verify the histogram $\{(0, 2^p)\}$. They are capacity measurements, each from one execution; there is no explicit-oracle timing or speed ratio for these enormous cuts. The serialized certificates are measured as sorted, uncompressed JSON using the project's exact integer codec. Decimal megabytes mean 10^6 bytes.

T	p	Produce (s)	Replay (s)	Total (s)	Certificate bytes	Decimal MB	P/a
1	16,384	0.008944	0.011613	0.020556	1,701,021	1.701021	0/0
4	1,024	0.360605	0.241919	0.602524	2,510,984	2.510984	15/3
4	16,384	1.224719	0.858719	2.083438	30,078,776	30.078776	15/3
8	16,384	6.404215	4.195131	10.599347	71,291,795	71.291795	47/3

Table 6: Separate native binary-capacity measurements. The last column lists ports P and maximal interval attachments a, rather than their numerical quotient. Serialization and hashing are outside the reported production/replay times.

The fixed number of exceptional interfaces does not imply a constant-sized proof. For example, the eight-tetrahedron case still has 47 ports and three interval attachments at $m = 2^{16384}$, but its full census certificate occupies 71.291795 decimal MB. Source coordinates and binary interval endpoints recur through the recorded proof traces. These are genuine output costs even though the 2^{16384} ball components are never individually materialized. The recorded SHA-256 digests identify the exact serialized proofs used in the capacity measurements.

7.6 Reproduction and the limits of the evidence

The complete per-case census is in `census-audit.json`; paired samples, arm order, A/A samples, capacity times, certificate lengths, and certificate digests are in `census-benchmark.json`. The diagnostic retains its per-case eligibility decisions in `annular-absorption-diagnostic.json`. From the repository's `fast/` directory, the supplied census drivers are invoked as follows:

```
python -m marked_parallelity_research.audit_census \
    --output census-audit.json
python -m marked_parallelity_research.benchmark_census \
    --output census-benchmark.json --rounds 5 --capacity
```

Regina is required for the explicit comparison arm. Native certificate production and replay use the repository's Python implementation.

The experimental conclusions have deliberately separate scopes. Passing tests and successful independent replay give evidence about implementation correctness. Agreement with independently constructed Regina cuts checks the topological interpretation on the manageable part of the corpus. Capacity runs demonstrate processing of large binary multiplicities and measure the resulting proof sizes. The all-input geometric and asymptotic claims rest on the theorems, with their stated source and interface hypotheses. Surface discovery, peripheral-data transport, complete hierarchy coverage, and a bound on the global recognition search remain outside these timings and outside the measured success count.

8 The remaining complexity gap and a concrete research programme

The proved results remove two local obstacles. We can retain the ordered attachment information needed for disk gluing without expanding a normal surface, and we can avoid a global high-dimensional intersection iterator when the matching cone has small coordinate factors. A complete recognition bound still needs a theorem connecting these operations to all knot inputs. This section specifies that missing theorem in measurable terms and proposes research tasks with clear success and failure conditions.

8.1 A ledger that prevents local and global bounds from being confused

There are at least four independent parameters in a hierarchy algorithm: the number of states visited, the encoding length of each state, the cost of finding a useful next surface, and the cost of verifying and updating its geometry. A source core of $O(T)$ handles controls only part of the last quantity, relative to the *current* source size T.

For example, suppose the next source has at most $30T$ exceptional handles. Iterating this estimate through h cuts permits $30^h T$ handles unless a separate contraction or amortization theorem intervenes. The bound does not remain $30T$ relative to the initial triangulation. Similarly, a sequence of polynomial-size individual normal coordinates need not be polynomially bounded in the original diagram if each size bound refers only to its immediate predecessor.

The following elementary accounting proposition is useful as a design gate. It is not asserted as a new general hierarchy theorem.

Proposition 8.1 (Controlled-growth search ledger)**.** *Consider a sound and complete deterministic recognition search represented by a rooted tree of height at most H. At depth i, suppose every state has at most S_i bits, every state offers at most S_i^a successors, and the overhead at each state, including terminal tests and successor enumeration, is at most S_i^b. In addition, the work per generated successor, including certificate handling, is at most S_i^b. Here $a, b \geq 0$ are fixed constants and $S_i \geq 2$. Then the total work is at most*

$$2(H+1)\exp\left(a\sum_{i=0}^{H-1}\log S_i + b\max_{0\leq i\leq H}\log S_i\right)$$

up to a polynomial initialization factor. If $S_i \leq A^i S_0$ for a fixed $A \geq 1$, this is

$$\exp\bigl(O((H+1)\log S_0 + H^2\log A + \log(H+1))\bigr).$$

In particular, $S_0 = n^{O(1)}$ and $H = O(\log n)$ give $\exp(O((\log n)^2)) = n^{O(\log n)}$.

Proof. At depth j, the number of nodes is at most $\prod_{i<j} S_i^a$. Each edge can be charged to its distinct child, so the numbers of nodes and edges are each bounded by $(H+1)\prod_{i<H} S_i^a$. Every node and edge charge is at most $\max_i S_i^b$, which gives the first estimate. Under the growth assumption, $\log S_i \leq \log S_0 + i\log A$. Summing this arithmetic progression proves the second estimate and the specialization. □

If every state remains polynomial in n independently of depth, height $O((\log n)^2)$ instead gives $\exp(O((\log n)^3))$. If only a fixed multiplicative state-growth bound is known at that height, the displayed ledger permits a fourth power of $\log n$. This difference explains why an explicit state-encoding bound matters even after a hierarchy-length estimate has been obtained.

Bounds on expanded cell counts can be much worse; the present work shows how to avoid them for a particular terminal operation.

For a fixed sector, the new candidate count $\sum_j \binom{h_j}{d_j-1}$ gives another useful parameter test. If $h_j \le n^c$ and $\max_j d_j \le C \log n$, each block has $n^{O(\log n)}$ candidate intersections, regardless of the sum of the d_j. That observation is a supplied-sector bound. Enumerating exponentially many sectors would negate it, and searching a whole tree with quasi-polynomial branching can increase the final logarithmic exponent. Candidate work and branching work must be charged separately.

8.2 Immediate geometric extension: absorb disk-base annuli

The annular absorption proposition in the geometry section is the most direct next implementation. It identifies a restricted annular interface that already has a complete topological interpretation: a disk-base bundle attached along its entire vertical boundary is one additional two-handle. Its raw co-core occurrences can be read from the authenticated core-one ports of its circular word. Appending these rows before cancellation can remove some refusals of the current consumer without importing a general annulus-recognition oracle.

The acceptance gate should require the whole component, not just a circle with a suggestive profile: connected disk base, one boundary component, product interval bundle, one full core circle, multiplicity one, and source verification of every crossing chart. The circle's signed walk can be checked against the ends of the core one-handles. For the source edge-cap face sector, the initial endpoint's tetrahedron and face flags determine which one-handle end is entered; both flags are needed for self-face gluings. Taking absolute occurrences after this geometric check supplies the saturated matrix row.

Research question 8.2 (Annular absorption coverage)**.** After this extension, how many actual source cuts become certifiable, and which residual annuli remain? Does absorption commute with the source leaf schedule in a useful canonical order, or does it create a genuinely larger positive-genus pivot search?

The first experiment should rerun the same 106 sources and report accepted cuts, independent replays, explicit-oracle agreements, and each remaining refusal category. The existing sixty annular refusals are a denominator for that experiment, not a claim that sixty cases will succeed after absorption. After the existing cancellations, the added attaching rows may have no singleton pivot, or the augmented core may not be a handlebody.

A separate source-profile diagnostic narrows the immediate target: in 38 of the sixty annular refusals, every full-circle attachment lies on a disk base; the other 22 have at least one nondisk-base circle. Saved profiles resolve 45 cases, and fifteen additional ordered queries, all independently replayed, resolve the rest. These 38 cases satisfy the absorption eligibility condition. They have not been reclassified by an augmented cancellation implementation.

8.3 General annular attachments need relative topology

An orientable interval bundle over an annulus is a solid torus. Gluing it to a core along one or more vertical annuli depends on how their core curves sit in the boundary of that core. Base Euler characteristic and an incidence graph do not encode this information. Even for a disk-base two-handle, the existence of a geometric singleton is stronger than an exponent sum condition.

Research question 8.3 (A small relative attachment language)**.** Can the source words be enriched by a polynomial-size representation of a meridian system and peripheral curves so that

primitive annular attachments, products, and selected handle cancellations are independently recognizable? Can this be done while retaining exact source embeddings through every update?

For an absorbed disk-base two-handle, a certified unique intersection with a retained geometric meridian supplies the cancellation proved by the disk-slide lemma. General annulus-base pieces still require relative gluing analysis. A wider class might admit a finite sequence of source-chart disk slides with compressed descriptions. An arbitrary free-group normalization should not be promoted to a geometric move unless its effect on the boundary embedding and the remaining attaching curves is also justified.

8.4 Transport boundary patterns through successful cancellations

The present certificate proves that a suitable disk system exists after the slides. It need not construct every disk or the induced boundary homeomorphism. This suffices for a handlebody genus census because arbitrary embedded disks in a handlebody boundary are legitimate gluing disks. It is insufficient for a next hierarchy step whose hypotheses refer to the exact position of a boundary pattern.

Research question 8.4 (Compressed pattern transport)**.** Is there a representation of curves and trivalent patterns on the core boundary that is closed under the meridian slides of the positive pivot lemma and has polynomial bit growth through a successful raw epoch? Which relative isotopies must a checker record to preserve pattern incidence and essentiality?

A useful certificate should carry source flags for each pattern strand, ordered contacts on its receiving chart, and enough local crossing data to distinguish disjoint arcs from equal homology classes. A directed acyclic description of transported strands is a candidate representation, but its size and the cost of intersection queries need proof. The cap-two semiring alone loses exactly the information such a transport requires.

8.5 Handle cores with three-handles

Ten of the saved corpus attempts stop at a core containing three-handles. The current refusal is intentional. An Euler calculation on the remaining handles does not decide how a three-handle closes a spherical boundary component, and simply deleting the corresponding row of a chain complex does not preserve the manifold.

Research question 8.5 (Source-certified two/three simplification)**.** Can an actual source two/three cancellation or a certified spherical-boundary reconstruction remove the three-handles while retaining the marked interfaces? Is there a geometric subfamily where its complete search has a polynomial certificate and producer bound?

A first implementation should authenticate the attaching sphere through its source vertex-link triangles and verify the geometric cancellation condition. One should separately investigate whether sphere components split off rather than cancel. This task must not substitute a homology-ball test for a ball test.

8.6 Reduce proof volume without weakening the checker

The large binary tests expose a practical cost that an asymptotic polynomial statement can hide: repeated source charts and long endpoint integers occur in several orbit certificates. The core proof is small, but the surrounding source and observer proofs can be tens of megabytes.

Research question 8.6 (Shared source proofs and persistent query systems)**.** Can the six certificates share a source packet, recurring interval maps, and unchanged trace fragments through a canonical directed acyclic representation, while independent replay still validates every query's distinct relation and weight transport?

The first target is serialization and repeated parsing, not a change in the mathematical result. Binary integers should have one canonical exact encoding; the existing hexadecimal transport avoids decimal conversion limits but is not a compact binary container. Any new schema should retain operation caps and reject source substitution, duplicate marks, unsupported trace events, and producer-supplied summaries without proof. The alternative three-count partition certificate in the interface section provides a different route whose actual proof size can be compared with sparse weight masks.

8.7 Measure and explain coordinate-block width

The finest matching decomposition is exact, but its usefulness depends on the sizes of the resulting blocks. The bipyramid family is deliberately transparent: it establishes a genuine asymptotic separation for the maintained iterator, but the separation theorem makes no claim that its positive-Euler witnesses are useful essential disks. It does not show that hard unknot exteriors have useful low-width witnesses.

Research question 8.7 (Geometric causes of small matching blocks)**.** Which topological or combinatorial decompositions force small matching blocks after triangle elimination? Which source moves preserve that property, and which make a single block absorb most quadrilateral coordinates?

The diagnostic should record maximum block nullity, total nullity, projected hyperplane counts, minimum-interaction cluster sizes, source vertex links, and the topology of the actual returned surfaces. Comparing only total nullity hides the improvement. Comparing only the smallest Q blocks hides the possible obstruction to full standard enumeration.

8.8 Treat Euler localization and useful-disk discovery separately

The Euler theorem remains exact even when triangle minima couple the Q blocks. That is a useful asymmetry: coupling does not hurt the Euler sign test, but it can create additional standard rays. The explicit two-ball counterexample proves that an implementation must keep these two APIs separate.

Research question 8.8 (Useful witnesses after an Euler-positive answer)**.** When can a block-local Euler witness be converted to a connected essential disk with a certified boundary slope or required relative pattern condition? When no such conversion succeeds, which further standard rays must be examined?

The correct experiment counts useful disks after all geometric predicates, not merely positive values of the Euler functional. The existing source component and boundary-essentiality machinery is an appropriate checker for that experiment. A negative canonical-Euler result, a positive but inessential disk, an incomplete search, and a verified essential disk should remain distinct outcomes.

8.9 The decisive missing search theorem: sector coverage

Research question 8.9 (A small complete family of candidate sectors)**.** Can every knot exterior at the relevant stage be supplied, in quasi-polynomial time, with a quasi-polynomial-

size family of compatible sectors containing one useful decomposing surface, with controlled local block or interaction dimensions? If this is false, what obstruction family shows its failure?

This is stronger than theorems about checking a supplied vector. A productive approach is to identify a progress property first and then formulate a structural existence theorem for surfaces satisfying it. Support-sensitive search, matching-sign implications, linear-programming witnesses, and the new block localization can all be candidate generators. None alone certifies that the family searched contains a useful surface on every input.

Possible falsification experiments should enumerate small genuine source triangulations, certify their topology independently, and compare the proposed restricted family against complete standard-ray searches where feasible. Abstract matrices are valuable for algebraic obstructions but cannot establish or refute a source-realized geometric existence claim on their own.

8.10 Admissible patterned states and uniform source size

The inherited core is a convenient source handle structure, not automatically an admissible patterned hierarchy state. The audited repository already contains an explicit empty-cut one-tetrahedron solid torus whose inherited characteristic-surface index violates the required admissibility condition. Thus bounded core size does not discharge that obligation, even on the simplest source fixture[6].

Research question 8.10 (A verified conversion to the hierarchy state model)**.** Can a source core, its ordered interfaces, and its transported boundary pattern be converted to an admissible structure of uniformly controlled type with a polynomial-size certificate? What potential decreases under the resulting geometric operations, and how does that potential bound both depth and restarts?

The recurrence in Proposition 8.1 should be filled with explicit functions derived from this conversion. It is insufficient to call every individual state "compressed." One needs a bound on the number of charts, the length of each affine or word description, and the accumulated bit length of ranks and markings after the allowed moves.

8.11 Formalize the small geometric bridge

Research question 8.11 (A manageable formal verification target)**.** Can the source chart-to-incidence lemma, the saturated positive pivot lemma, and the disk-gluing genus theorem be formalized as a small verified kernel, with machine-checkable hypotheses for an actual cut packet?

The finite arithmetic portions are especially suitable: exact integer codecs, spanning forests, source leaf orders, permanent-one implications used only in the meta-proof, and multigraph cycle ranks. The topological interface is more substantial. It must state that the source packet represents an embedded interval bundle and a genuine handle structure with the asserted intersection, not merely a cell complex having the same homology. Isolating that interface would make both the mathematical proof and future code review more reliable.

8.12 Evaluation and integration priorities

The next measured comparison should keep three axes visible: source complexity, coordinate bit length, and useful geometric coverage. A benchmark consisting only of enormous multiples

of a known disk establishes compressed capacity but does not measure hard surface discovery. Conversely, a bank of tiny primitive vectors conceals the asymptotic cost of explicit expansion. Both are needed.

The recommended integration order is:

1. Preserve this additive operator and its independent source-bound checker; keep the existing recognizer's decision contract intact.
2. Implement and audit the proved disk-base annular absorption rule.
3. Reduce repeated source/proof serialization and measure memory as well as producer and verifier time.
4. Enable block decomposition in the matching query paths whose ray-set or Euler contract it preserves exactly, measuring preparation and replay costs.
5. Build the relative pattern transport and useful-surface coverage theorems needed by an actual hierarchy consumer.
6. Establish a global state and search ledger in the original diagram size before claiming a quasi-polynomial recognizer.

The mathematical deliverable is useful even if the last step remains difficult: the new certificates identify precisely which topological information a compressed cut consumer can use, and the new counterexamples identify which apparently plausible shortcuts would lose necessary information.

9 Artifact layout, public APIs, and reproduction

The accompanying archive contains the complete article sources and PDF, an additive integration patch, a runnable source snapshot, fixed source corpora, proof examples, experiment records, and validation logs. The snapshot is included so that local reproduction does not depend on the current state of the repository's main branch. The patch is separately based on the exact revision printed on the title page.

9.1 Delivered files and their roles

Path	Purpose
`article.tex`, `sections/`	Complete editable article sources; the bibliography is inside the main source.
`article.pdf`, `build.sh`	Typeset article and local rebuilding command.
`integration.patch`	New modules, tests, and reproduction data for the pinned repository tree; no existing runtime file is overwritten.
`runtime/fast/`	Standard-library native implementation, its pinned dependencies, four new test modules, and fixture/benchmark tools.
`runtime/synthesis/`, `runtime/reports/57/`	Source corpora needed by the literal boundary and retained-sector audits.
`examples/`	Small independently verified source-bound proof packets, including the Q/standard mixed-ray example.
`results/`	Case-level audit and timing JSON, source hashes, and logs.
`tools/reproduce.py`	Portable smoke, focused-test, audit, and benchmark entry points.
`provenance/`	Pinned revision, file hashes, environment, patch replay checks, and PDF verification record.

The native runtime requires Python and its standard library. The recorded independent explicit-cut comparison uses Regina 7.4. A machine without Regina can run the new native

tests and replay the supplied small certificates; the explicit comparison mode requires Regina. Rebuilding the paper requires a LaTeX installation with the packages named in `article.tex`. The included upstream runtime license is preserved.

9.2 A source-bound whole-cut example

From `runtime/fast/`, the following computes and independently checks the cut along a supplied primitive meridian:

```
from fastunknot.normal_cut_handlebodies import normal_cut_handlebodies
from fastunknot.normal_cut_handlebodies_verify import (
    verify_normal_cut_handlebodies,
)
from normal_orbit_research.fixtures import layered_torus

source, coordinates = layered_torus(4)
answer = normal_cut_handlebodies(
    source, coordinates, record_certificate=True,
)
assert answer['status'] == 'COMPLETE'
assert verify_normal_cut_handlebodies(
    source, coordinates, answer['certificate'],
)
assert answer['census']['histogram'] == [
    {'genus': 0, 'multiplicity': 1},
]
skeleton = answer['certificate']['boundary']['skeleton']
assert len(skeleton['ports']) == 15
assert skeleton['interval_attachments'] == 3
```

This is a theorem about the supplied cut. Converting it to a knot verdict requires the additional surface and exterior hypotheses described above. An unresolved result should be passed to a different exact method or retained as unresolved; it must not be converted to a negative recognition answer.

For a large common multiplicity, multiply the normal coordinates by the exact integer. The native algorithms consume those binary integers directly. When saving a large certificate in JSON, use the established exact codec:

```
import json
from fastunknot.integer_codec import json_safe

source, primitive = layered_torus(1)
m = 1 << 16384
coordinates = [[m * n for n in row] for row in primitive]
answer = normal_cut_handlebodies(
    source, coordinates, record_certificate=True,
)
assert answer['census']['ball_components'] == m
wire = json.dumps(json_safe(answer['certificate']))
proof = json.loads(wire)
assert verify_normal_cut_handlebodies(
    source, json_safe(coordinates), proof,
)
```

The codec uses explicitly signed hexadecimal strings for sufficiently large integer fields,

avoiding process-wide decimal-conversion changes. Equal Boolean and integer values are not interchangeable schema fields. The checker validates the prescribed types and source correspondence; successful JSON parsing is not itself certificate validation.

9.3 A sector example that must retain the mixed standard ray

```
from sector_block_research.fixtures import capped_bipyramid_chain
from fastunknot.sector_blocks import certify_sector_block_rays
from fastunknot.sector_blocks_verify import verify_block_ray_certificate

fixture = capped_bipyramid_chain(2)
source = fixture['triangulation']
support = fixture['allowed_types']
q = certify_sector_block_rays(source, support, phase='quadrilateral')
s = certify_sector_block_rays(source, support, phase='standard')
assert len(q['rays']) == 4
assert len(s['rays']) == 5
assert verify_block_ray_certificate(source, q['certificate'])
assert verify_block_ray_certificate(source, s['certificate'])
```

The source and support are the same. The difference in ray counts is required by the proved canonical-minimum interaction, rather than a tolerance or a heuristic enumeration choice. The negative canonical-Euler and positive canonical-Euler certificates have separate schema names and weaker topological meaning than an essential-disk certificate.

9.4 Commands and scope of the validation records

From the archive root:

```
python tools/reproduce.py smoke
python tools/reproduce.py tests
python tools/reproduce.py audits
python tools/reproduce.py benchmarks
bash build.sh
```

The smoke command verifies the saved small proof packets. The test command runs the four new test modules; it does not claim to recreate all 1,644 tests of the integrated source tree. The audits use the included fixed source corpora and write new reports under `reproduced/`. Benchmark commands run the recorded algorithms with their original sample counts; wall-clock times will depend on the host. The binary capacity cases report exact arithmetic and certificate volume, not an explicit-oracle speed ratio.

To reproduce the full integrated suite, check out the pinned repository, inspect and apply `integration.patch` at the repository root, enter `Topology/UnknotRecognition/fast/`, and run:

```
python -m unittest discover -s tests -v
```

The delivered log records the actual 1,644-test run. A subsequent focused sector run validates the final generalized fixture as well. The separate runtime snapshot is sufficient for this contribution's focused tests and audits; it intentionally does not duplicate every unrelated report in the large ProveIt repository.

The integration patch only creates new files. It does not enable a new diagram recognizer

policy, replace existing source checkers, or assume that all future normal cuts will satisfy the disk-interface criterion. Its purpose is to make the proven local operations and their limits reviewable before a hierarchy consumer depends on them.

9.5 Results that are proved but not implemented in this delivery

Four extensions are deliberately separated from the measured code: the three-unweighted-count partition certificate, reuse of a primitive certificate under the replication theorem, exhaustive bounded-genus survivor search, and disk-base full-annulus absorption. Their mathematical statements and proofs are in the article. The current production APIs do not silently claim their speed, coverage, or proof format. The annular eligibility diagnostic tests only the stated base-type preconditions.

This distinction makes a later implementation comparison precise. A future patch can add one extension, replay the same source bank, and identify exactly which proof language, accepted cut class, and measured work changed.

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